

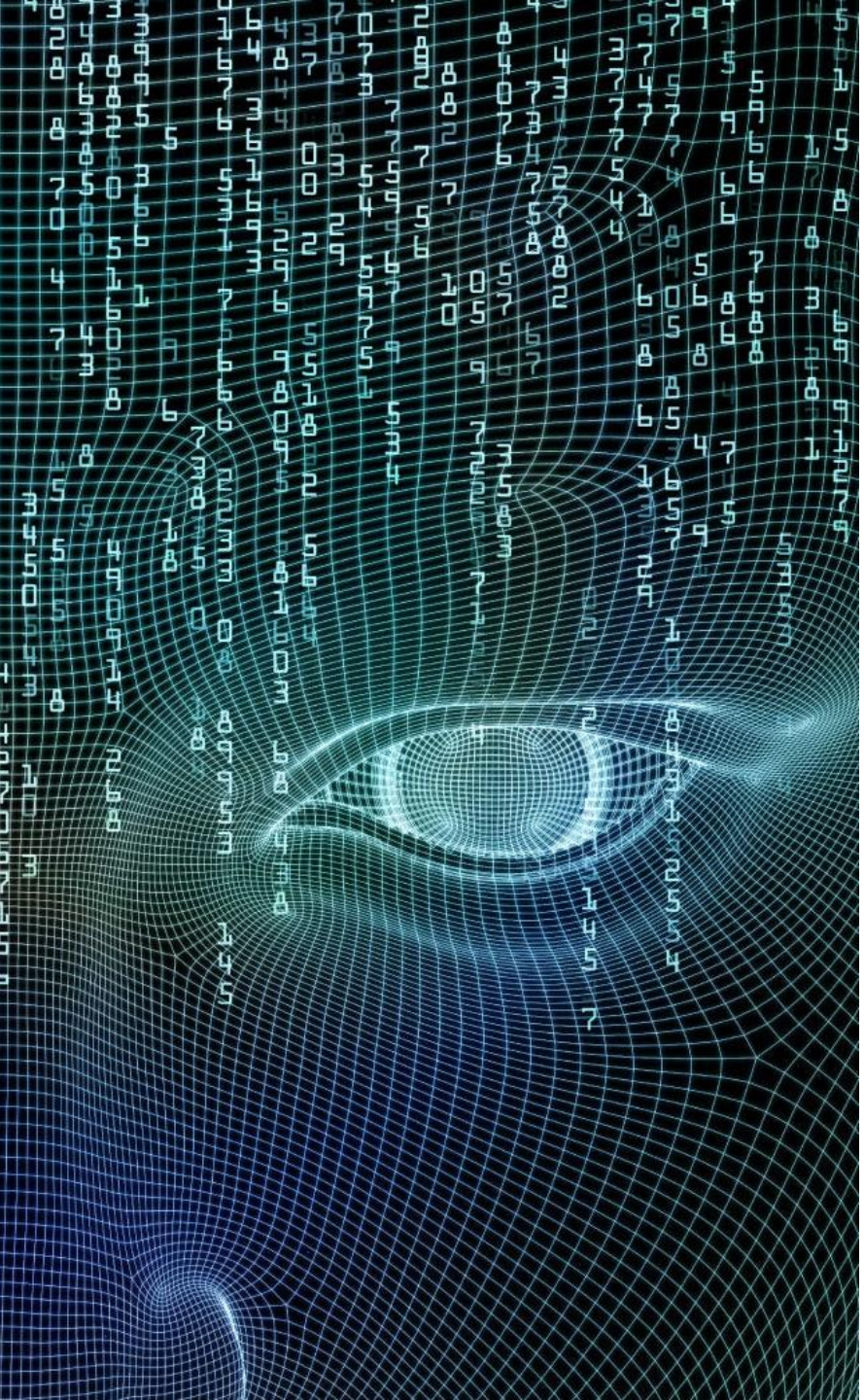
Spring of Code

Artificial Intelligence

(Machine Learning & Deep Learning)

Instructor
Asad Ullah
- *Research Assistant* (DIP Lab)

Duration
03 Months
(Feb – April)



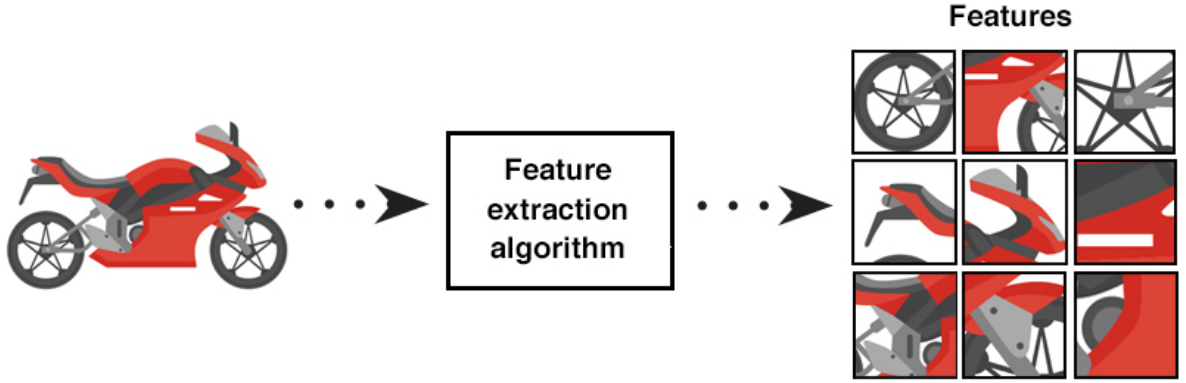
Day 01 – Deep Learning
(Convolutional Neural Networks)

Objectives:

- Image Features
- Convolution
- Pooling
- Convolutional Neural Networks

What Are Features in Images?

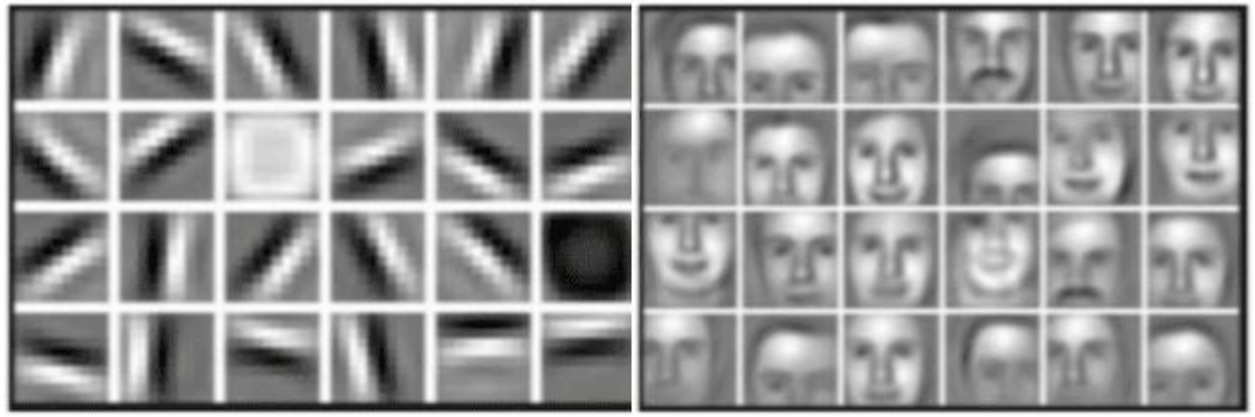
- What makes this image recognizable?
- Features are distinctive patterns or characteristics
- Examples: edges, corners, textures, shapes, colors that help identify objects





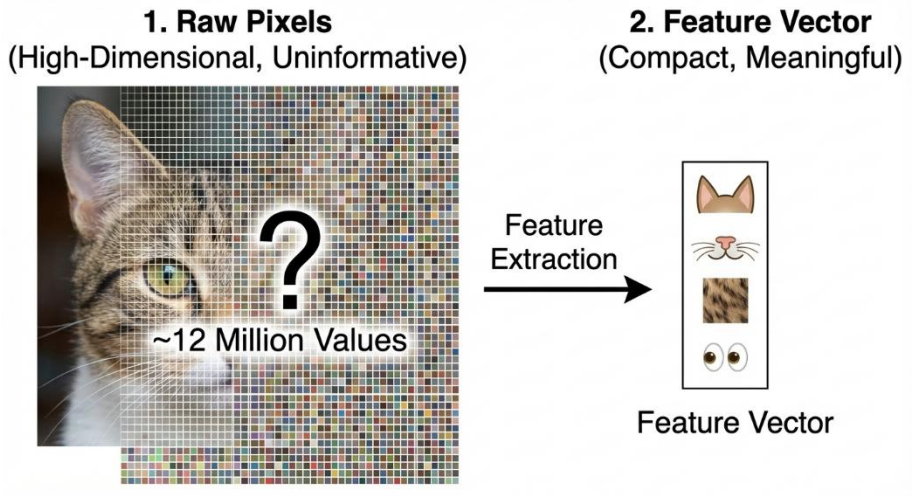


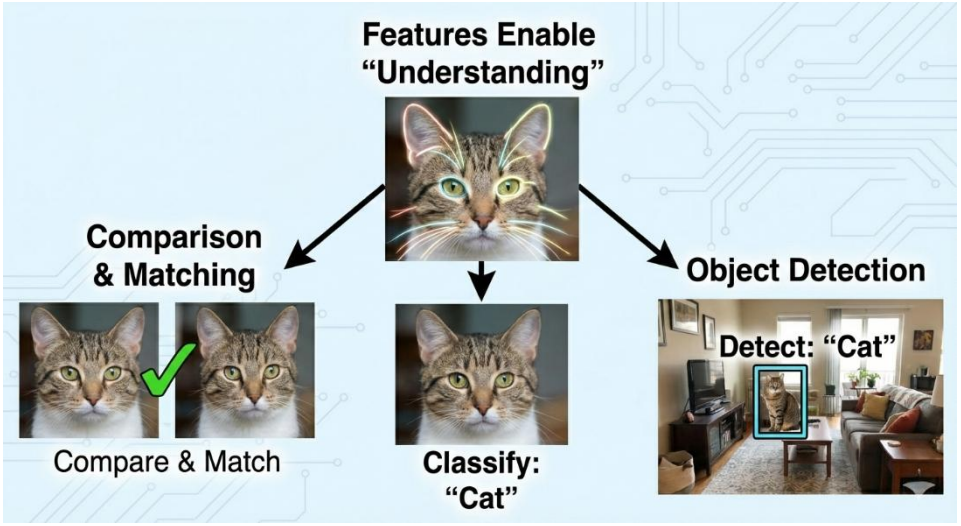




Why Do We Need Features?

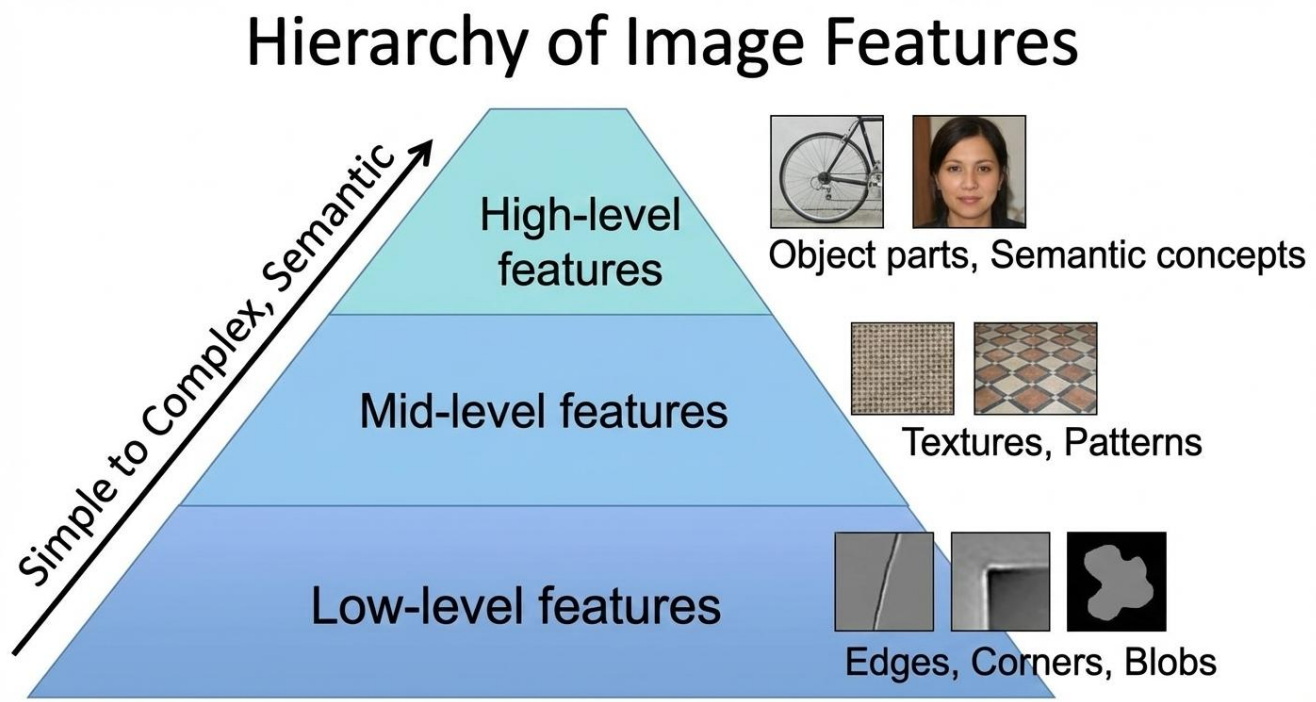
- Raw pixels are too high-dimensional and not informative
- Features provide compact, meaningful representations
- Enable computers to "understand" and compare images
- Foundation for image classification, object detection, matching

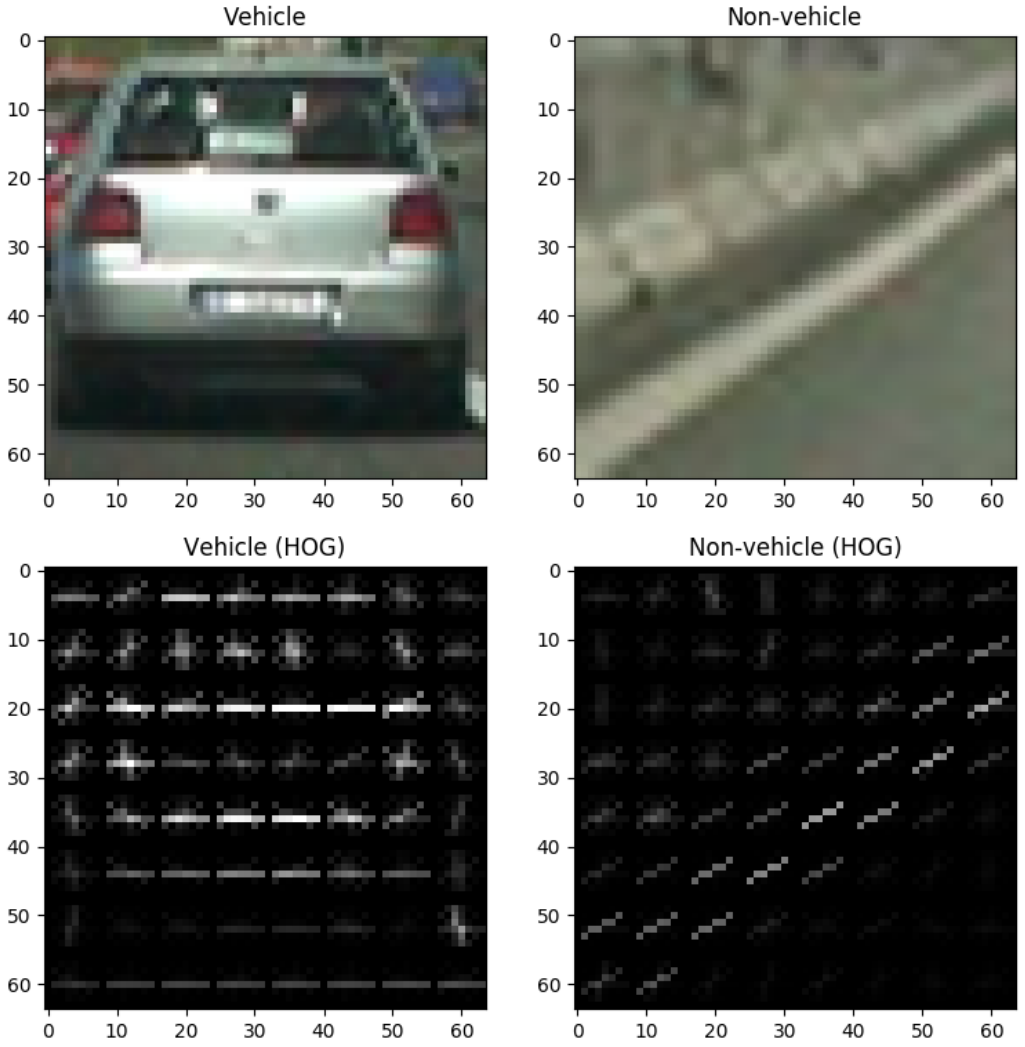


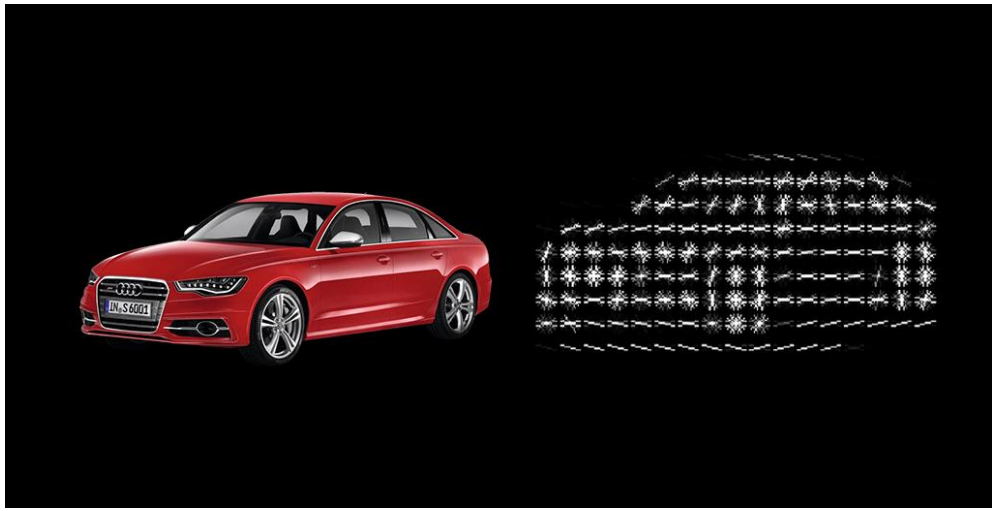


Types of Features

- **Low-level features:** edges, corners, blobs
- **Mid-level features:** textures, patterns
- **High-level features:** object parts, semantic concepts



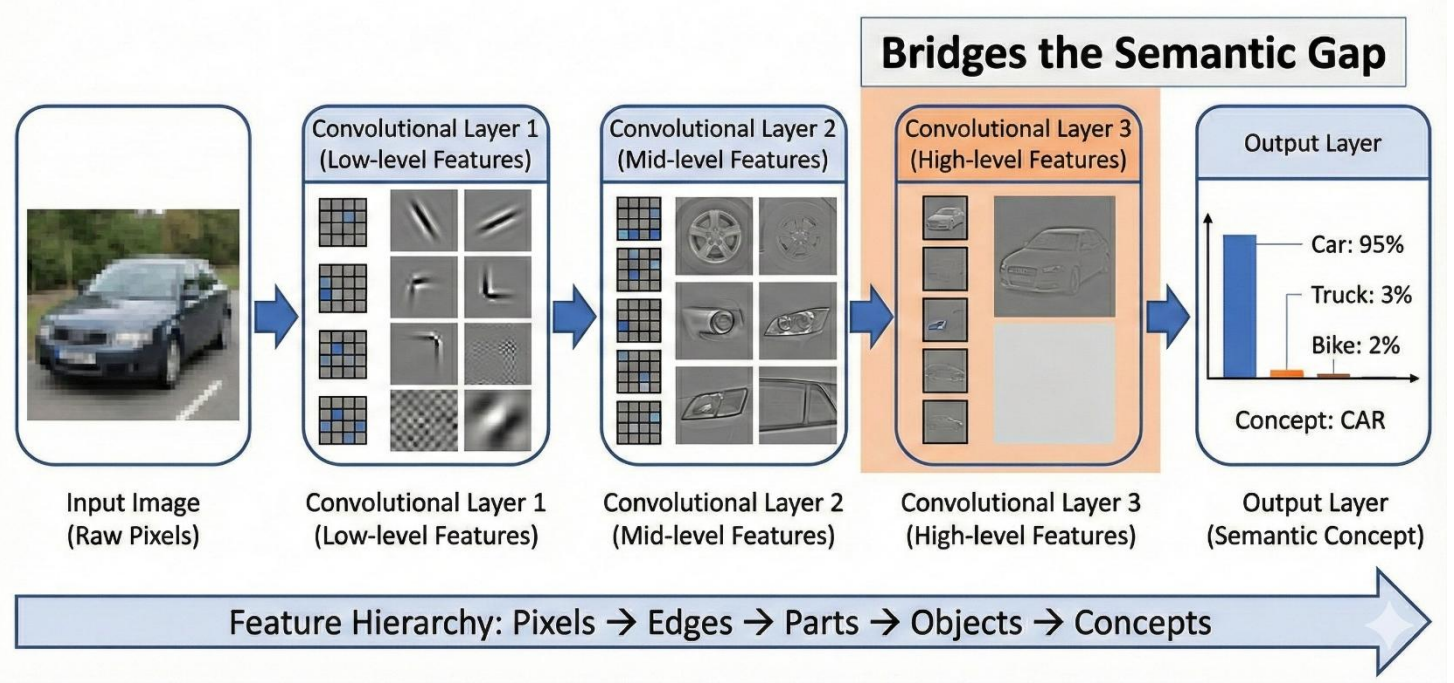

Vehicle
Non-vehicle
Vehicle (HOG)
Non-vehicle (HOG)

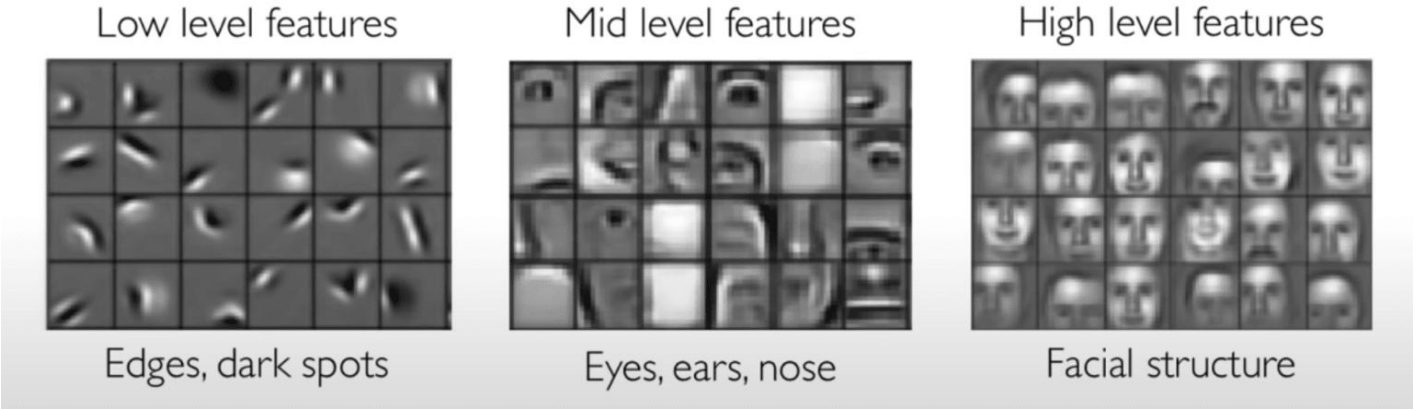



InputImage
LBPFeatures

Enter Deep Learning - CNNs

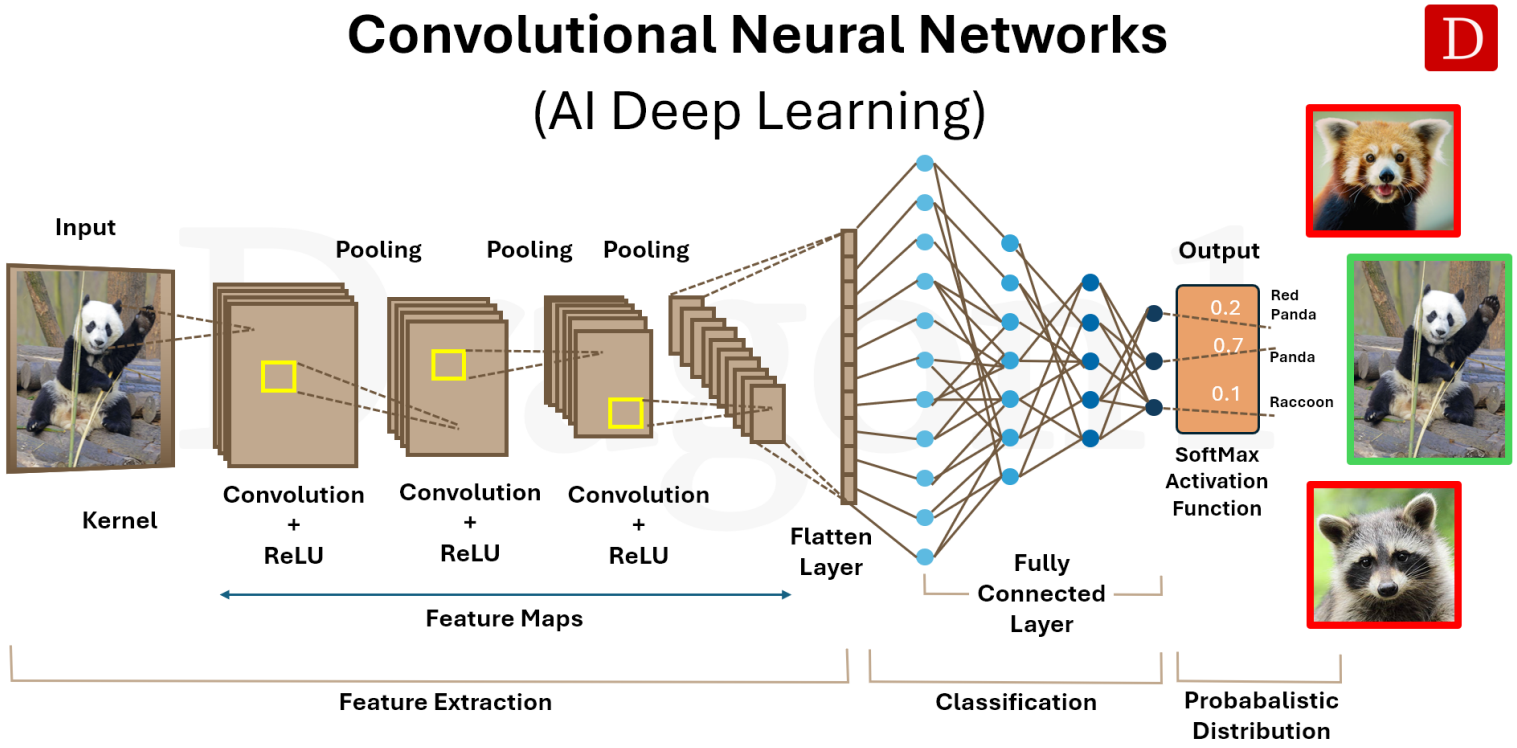
- Convolutional Neural Networks (CNNs)
- Learn features automatically from data
- Build hierarchical representations
- Bridge the semantic gap
- From pixels → edges → parts → objects → concepts





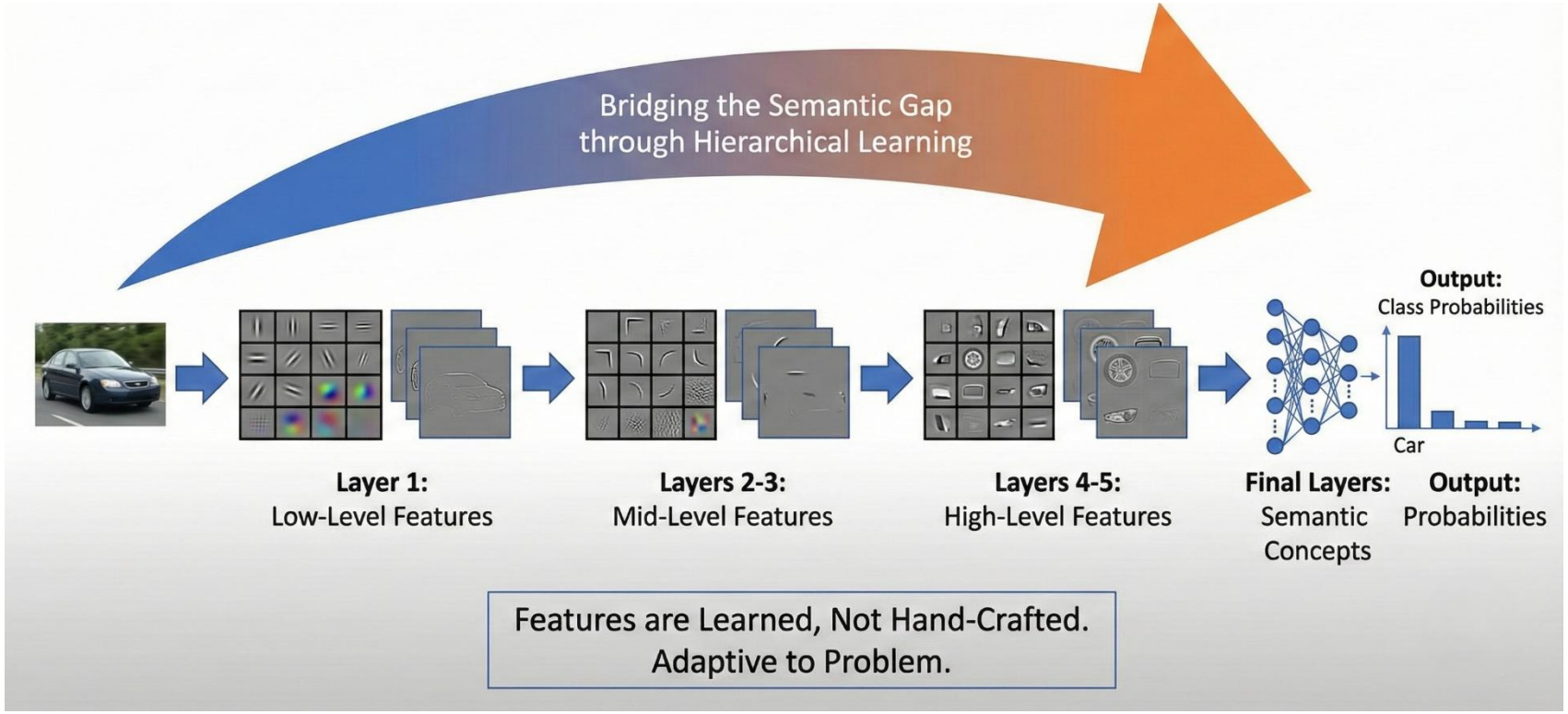
CNN Architecture Overview

- **Convolutional Layers:** Learn spatial features
- **Pooling Layers:** Reduce dimensionality
- **Activation Functions:** Introduce non-linearity (ReLU)
- **Fully Connected Layers:** Classification
- **Softmax:** Output probabilities



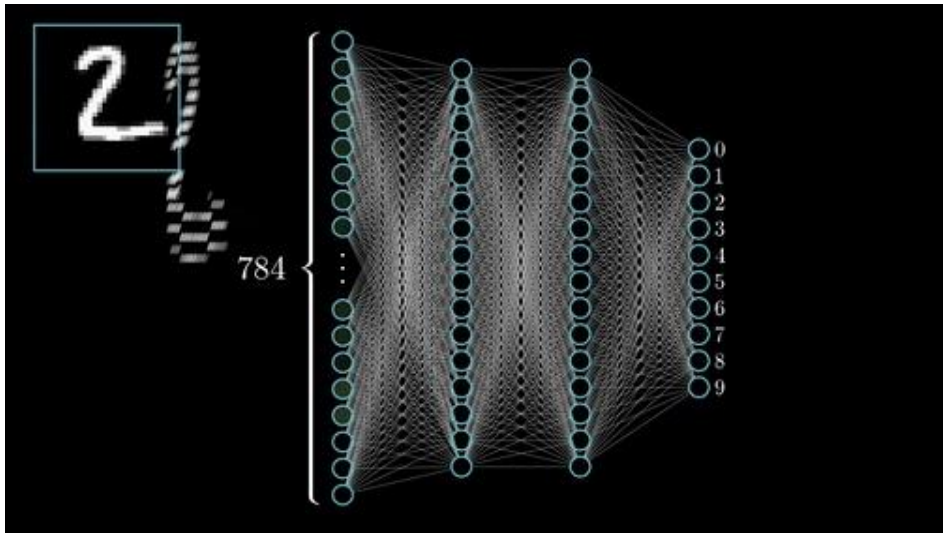
How CNNs Bridge the Semantic Gap

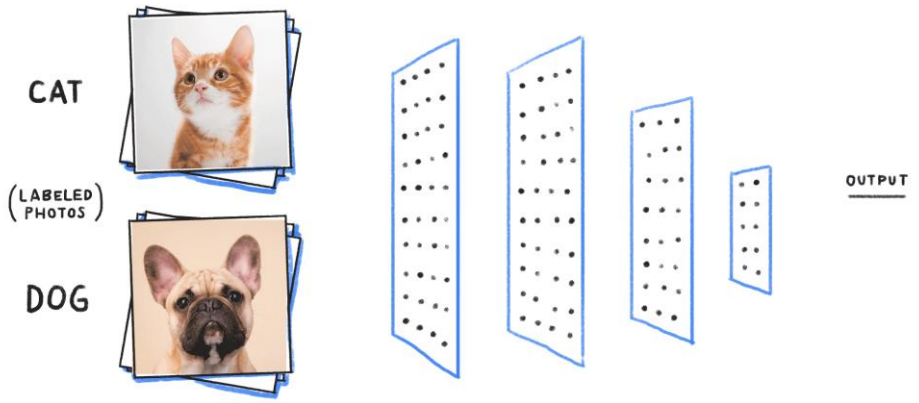
- **Layer 1:** Low-level features (edges, colors)
- **Layer 2-3:** Mid-level features (textures, patterns)
- **Layer 4-5:** High-level features (object parts)
- **Final layers:** Semantic concepts



Convolutional vs Fully Connected Layers

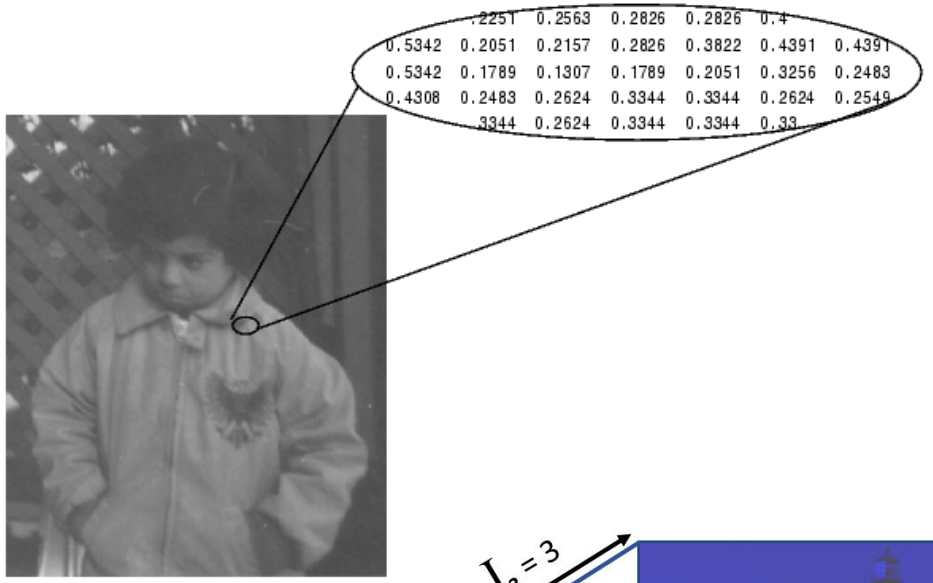
- Each neuron connected to a small range of input, instead of all the inputs.
- Performs the convolution operation to extract features.
- It has fewer parameters compared to fully connected layers because of weight sharing.
- Primarily used for feature extraction

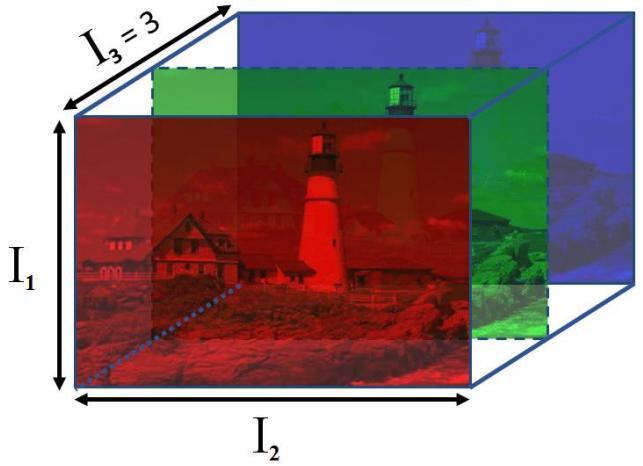


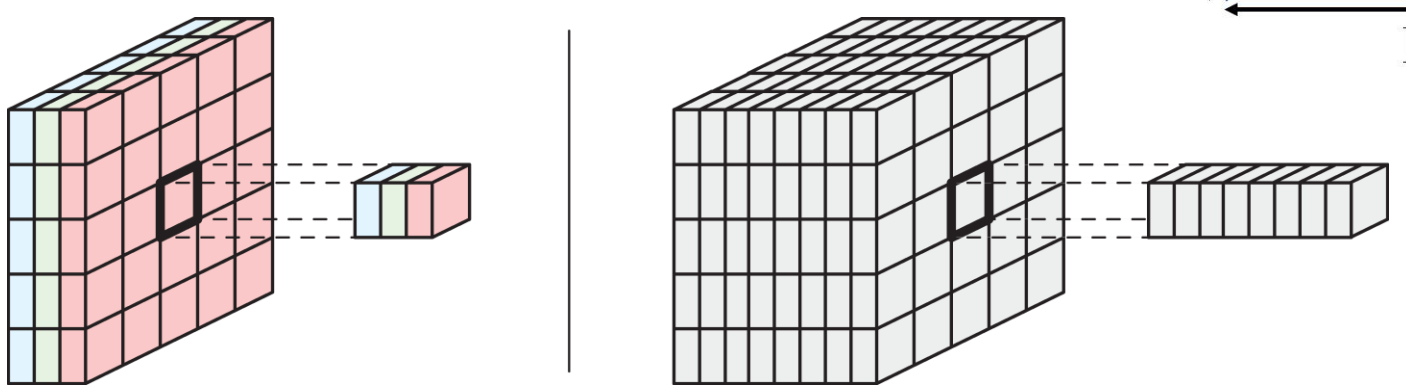


Images and Tensors

- A digital image is a grid of pixel values, each pixel containing intensity information.
- Grayscale images are represented as a 2D matrix of pixels (Height x Width), while color images have channels as well (Height, Width, Channels).
- Tensors are generalization of matrices to higher dimensions.







Convolution

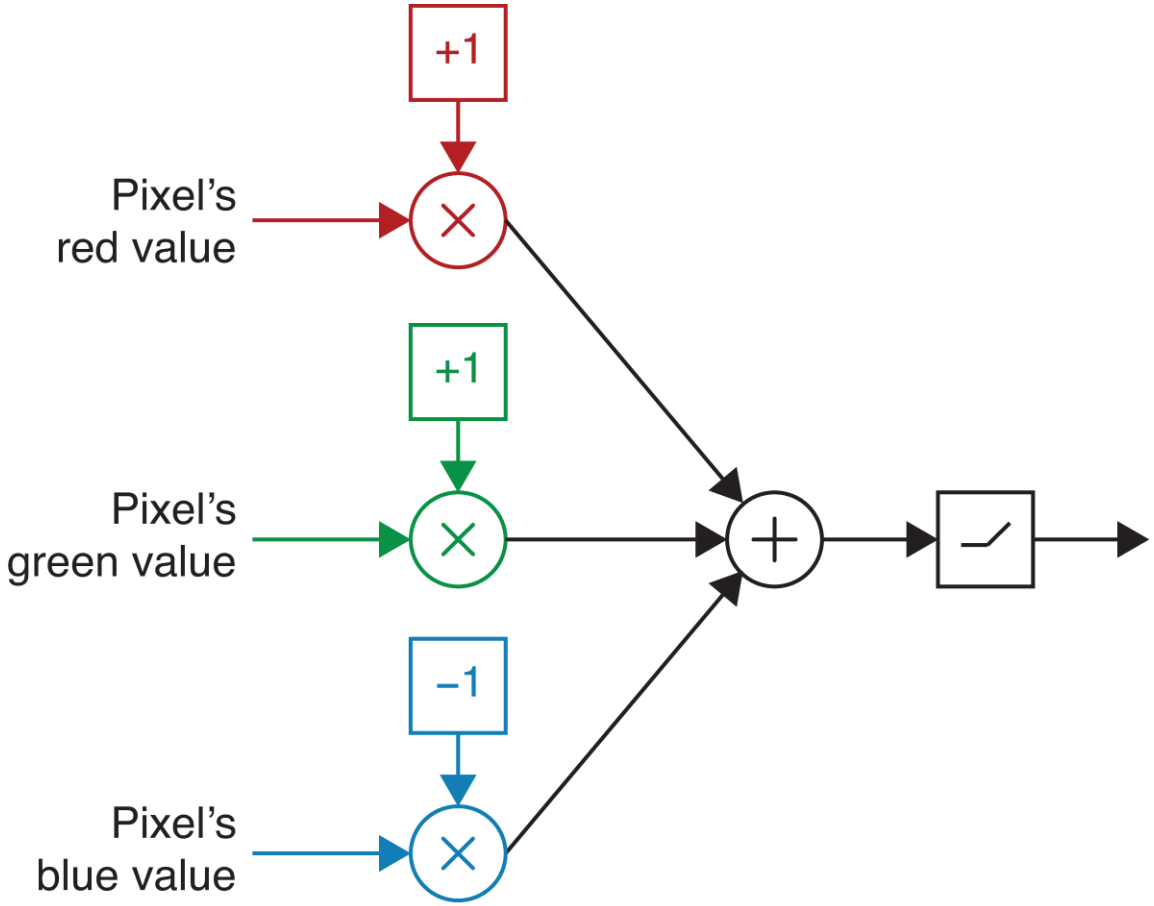
Let’s say we want to detect the intensity of yellow color in each pixel





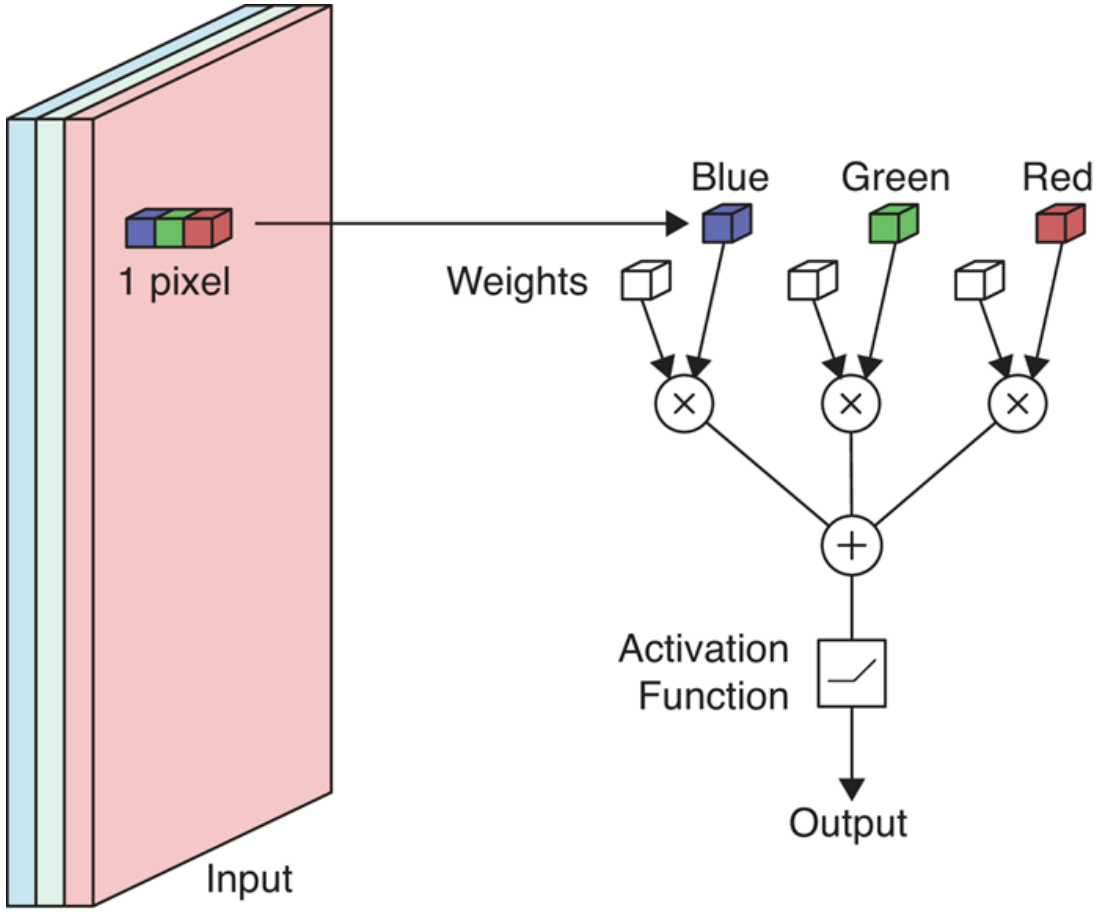
Convolution

We can create a yellow detector as an artificial neuron like so



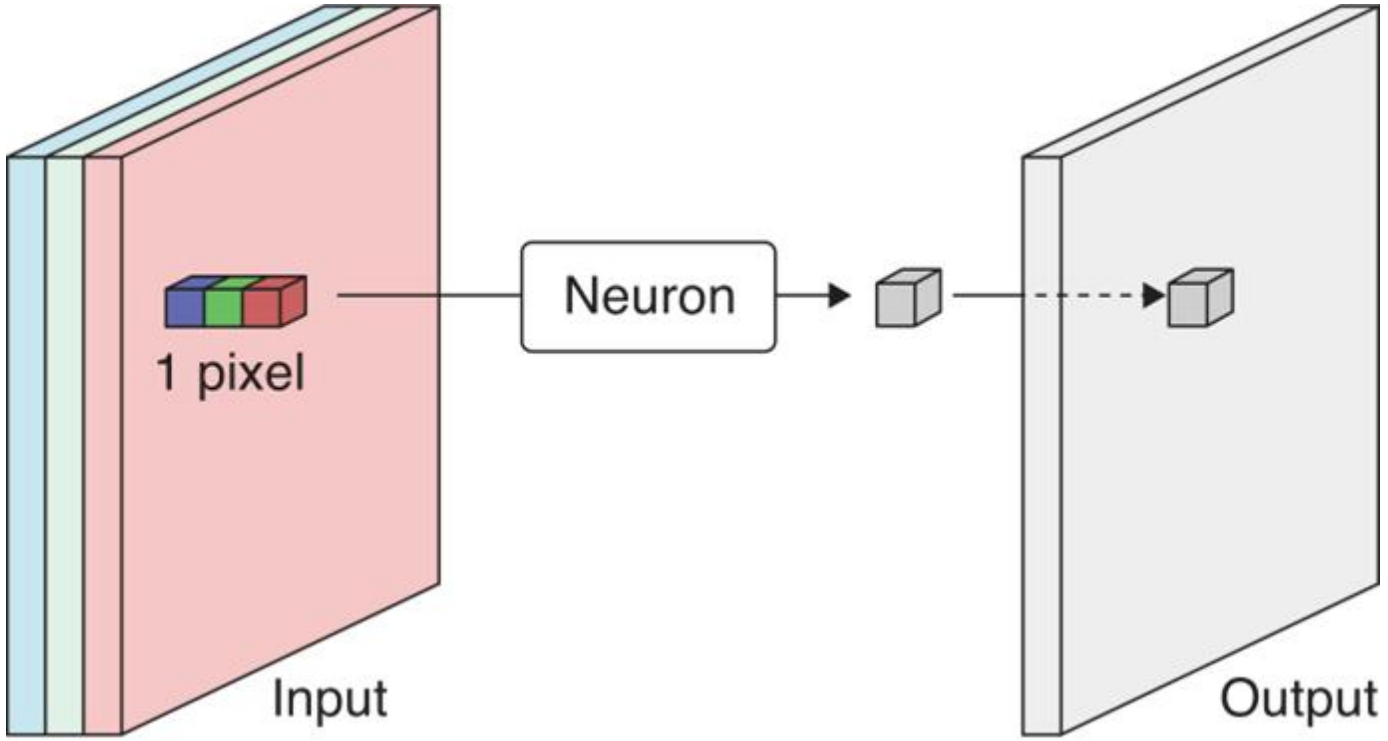
Convolution

We can apply the yellow detector to every pixel in the color input image



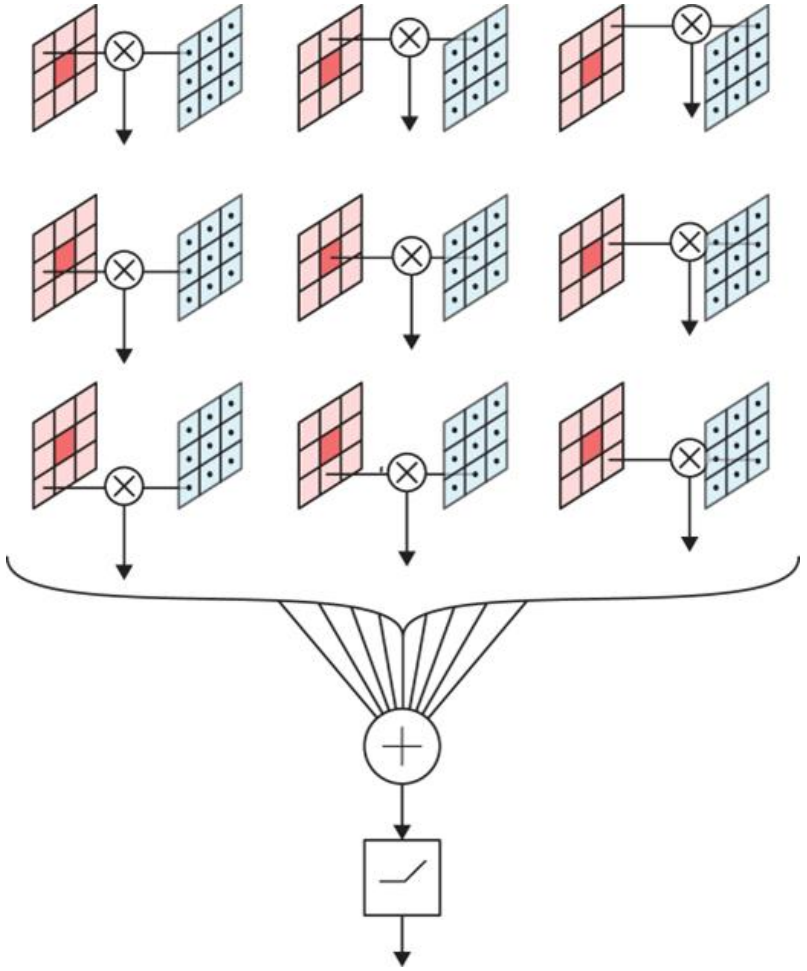
Convolution

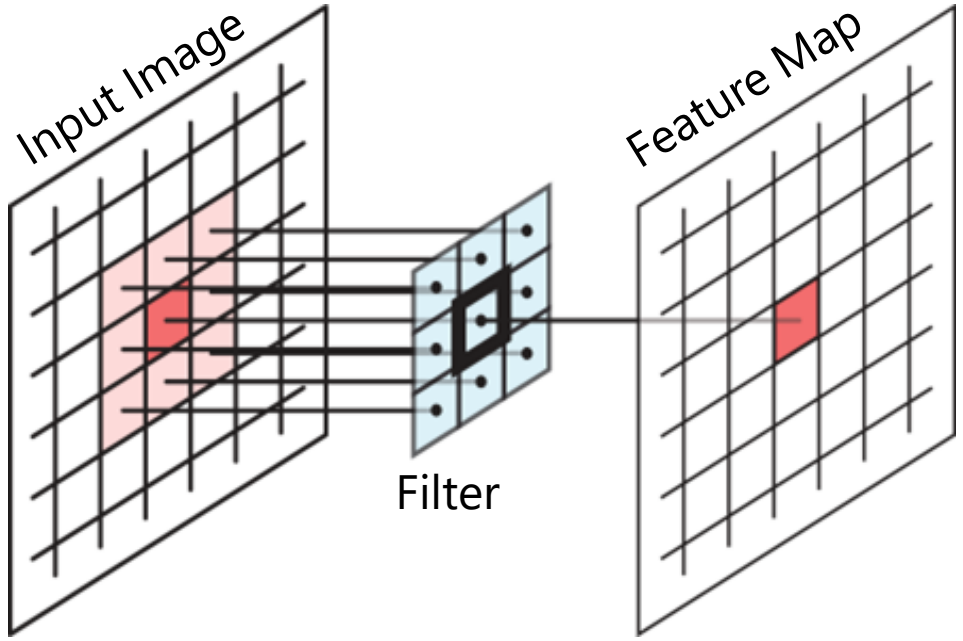
As a result, we get a grayscale image where each pixel corresponds to the intensity of yellow color in the input image



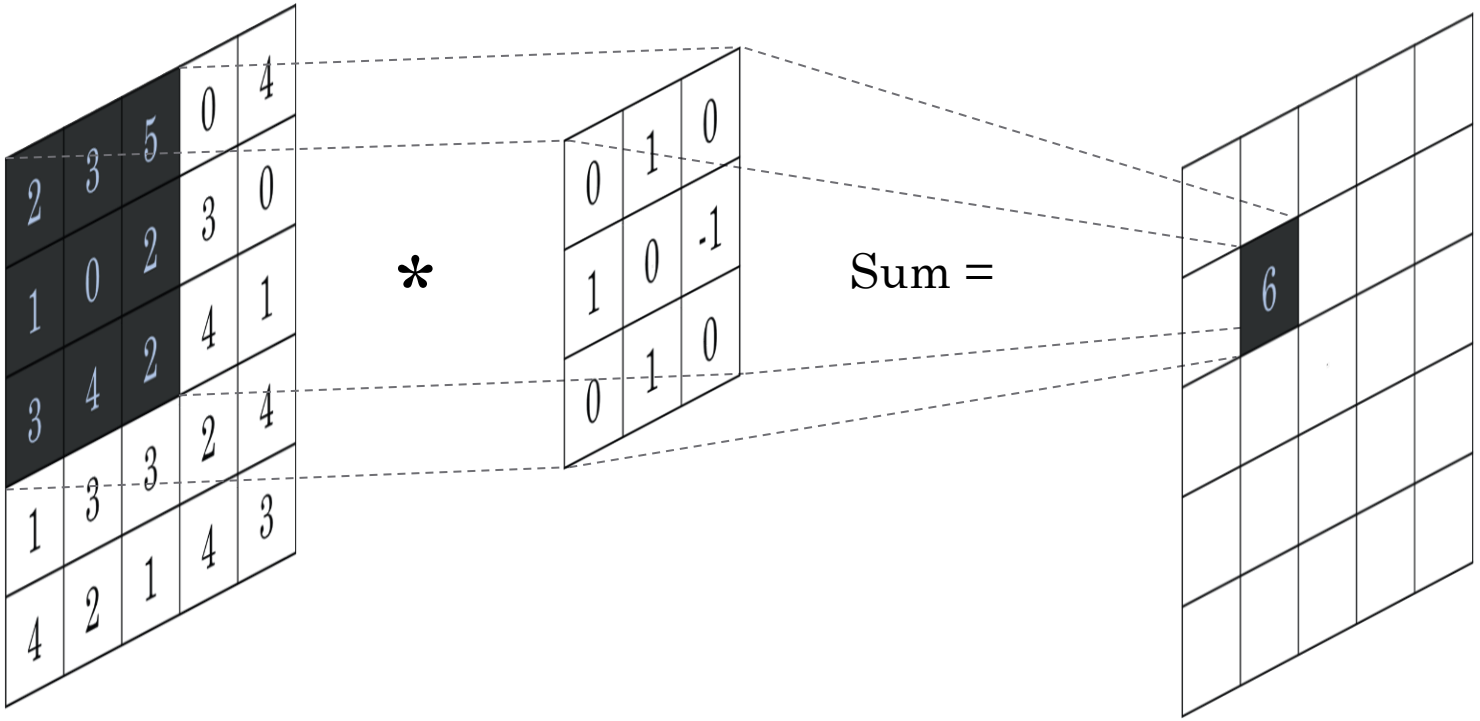
Convolution

Instead of processing a single pixel at a time, we can process a small region around the pixel



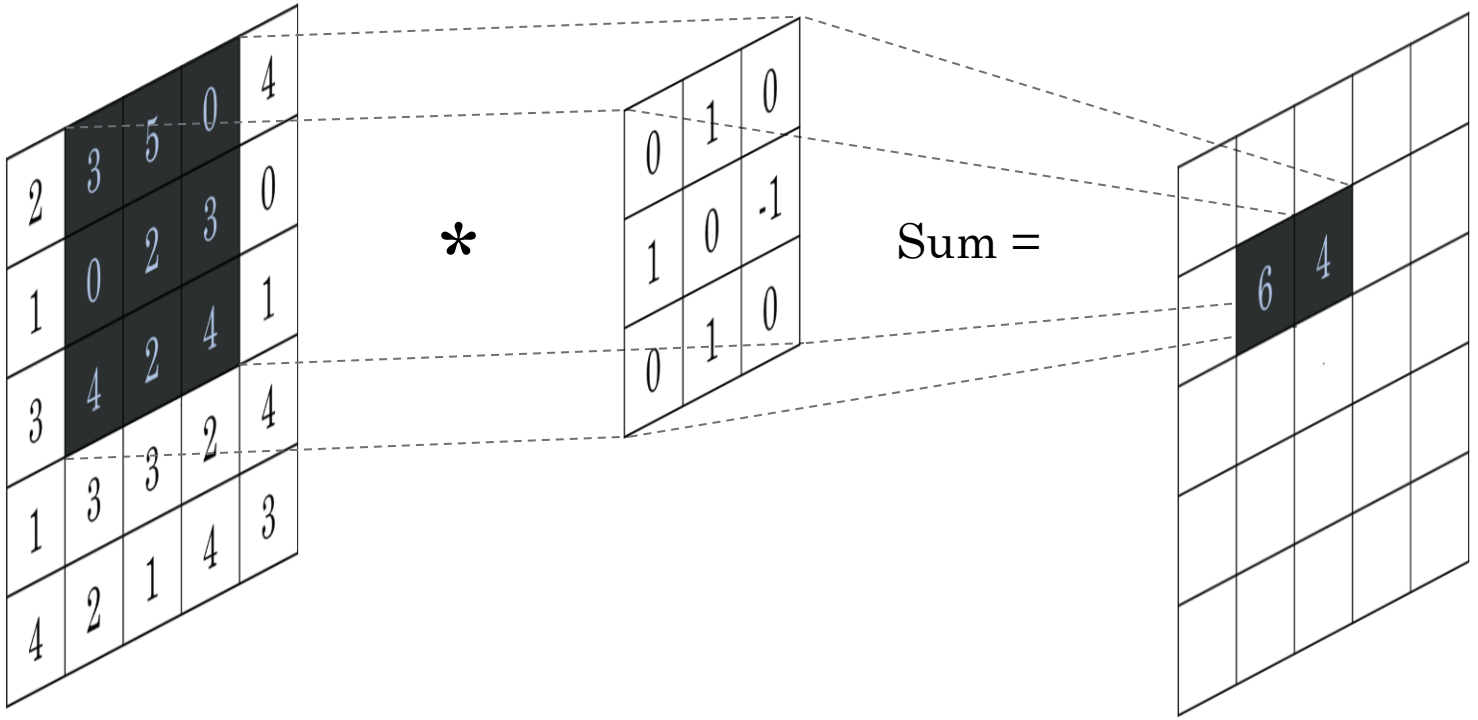


Example of Convolution Operation

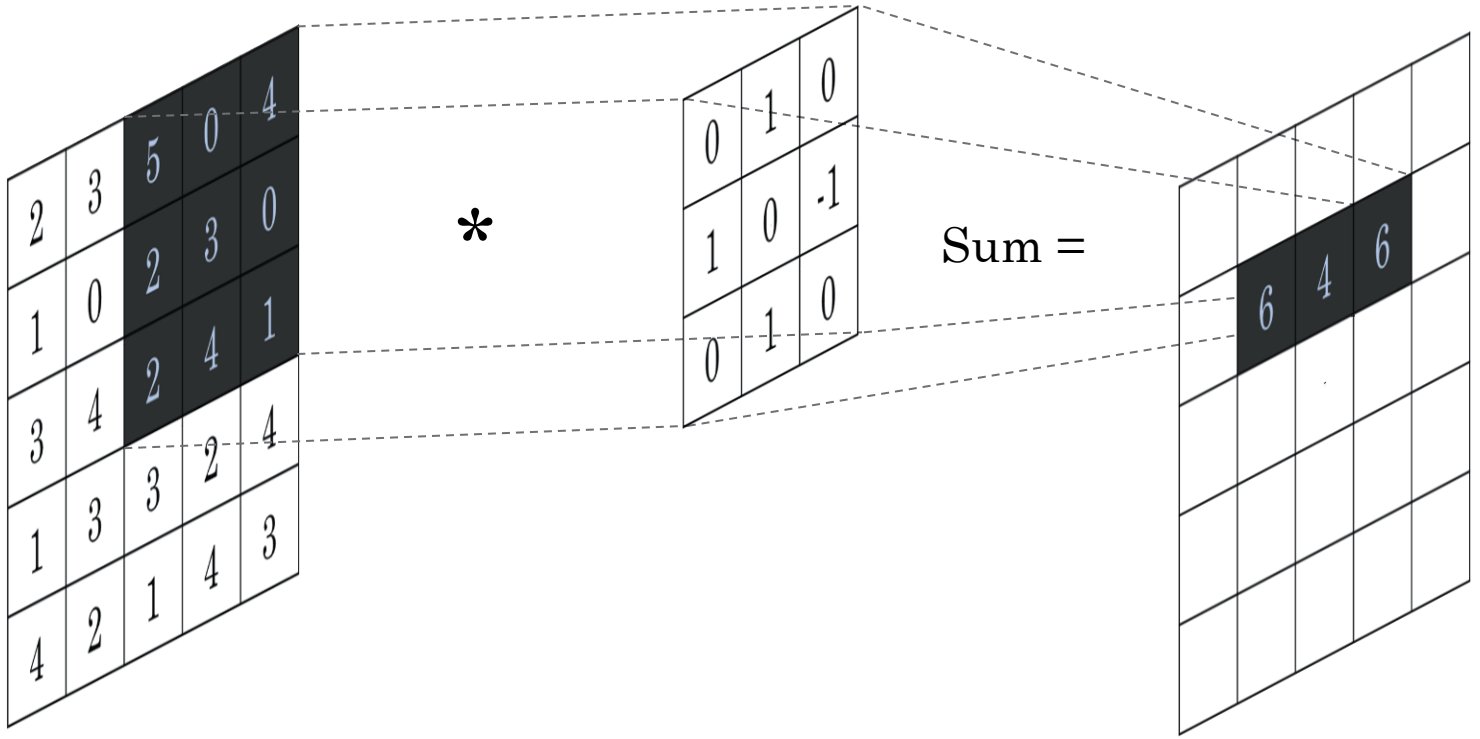


Input Image Filter Output Image

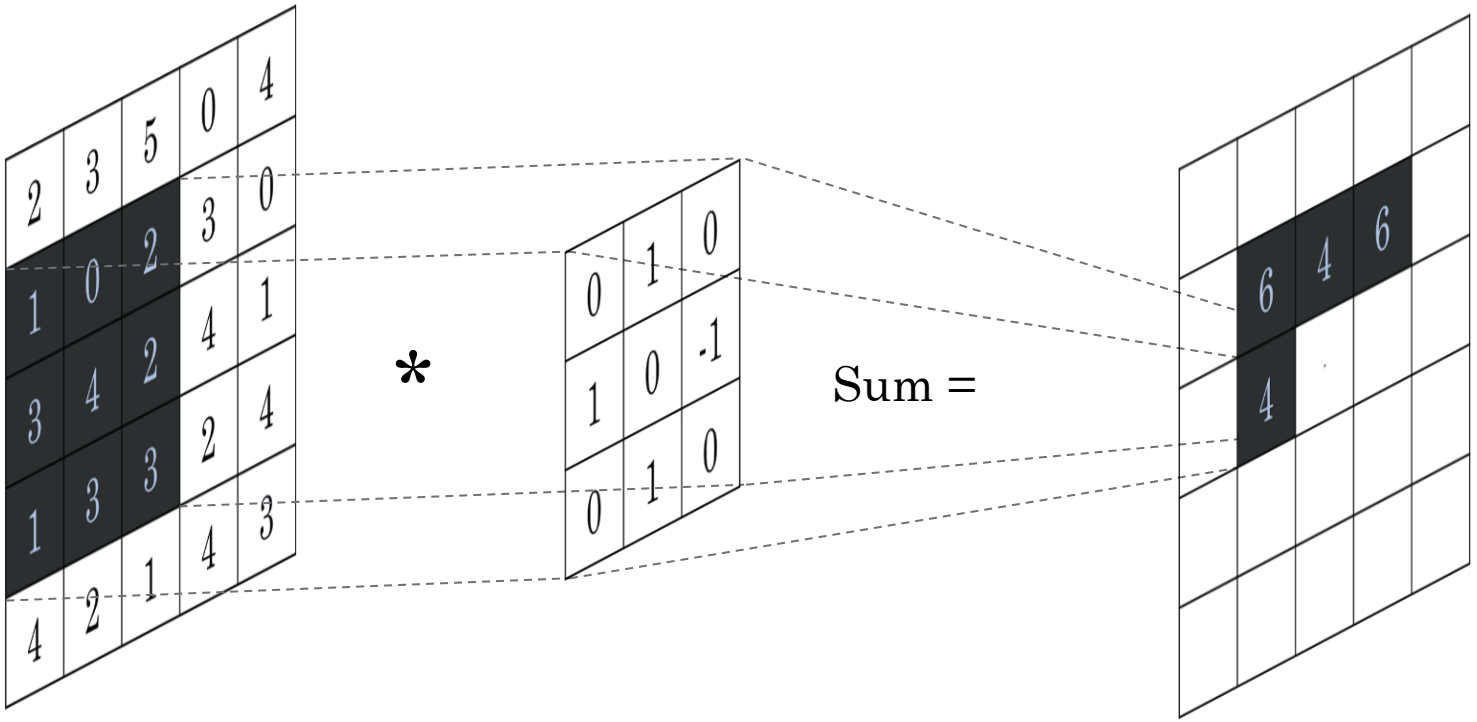
Example of Convolution Operation



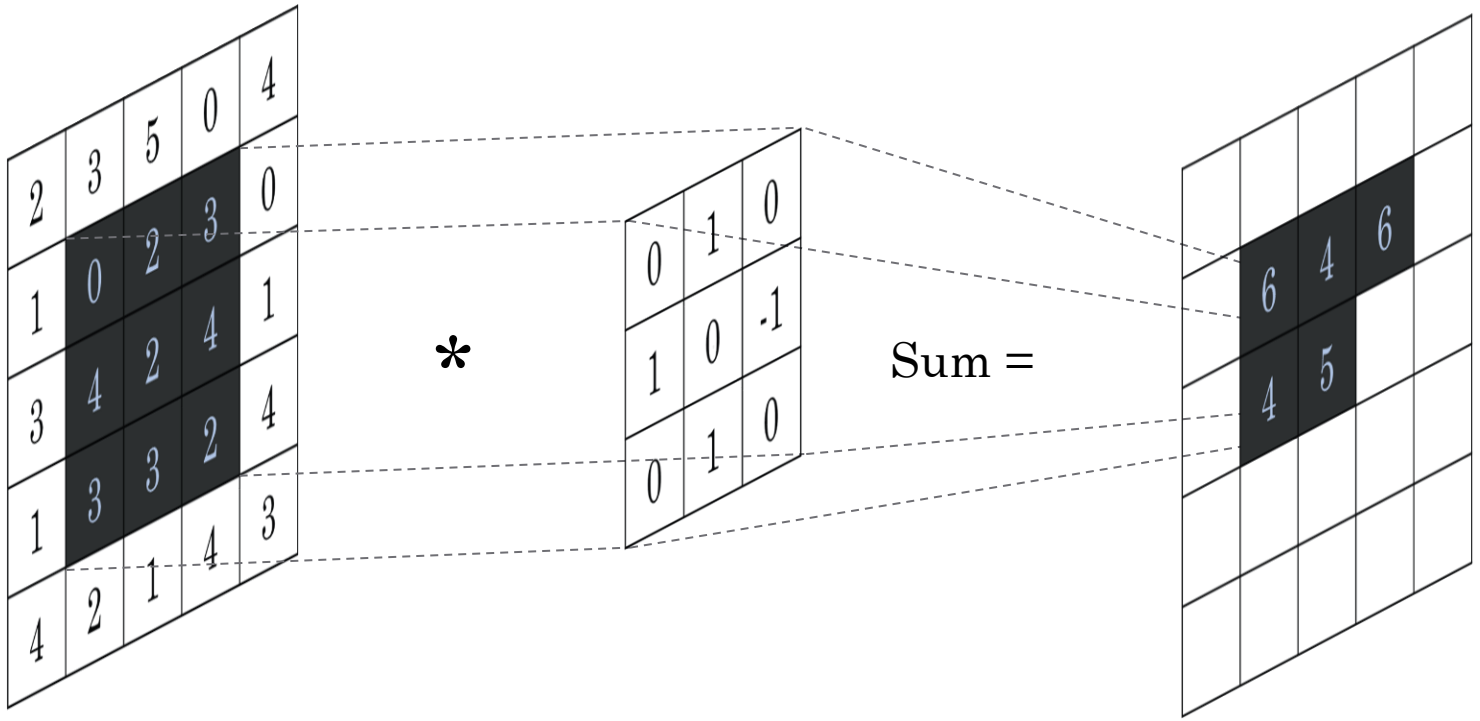
Example of Convolution Operation



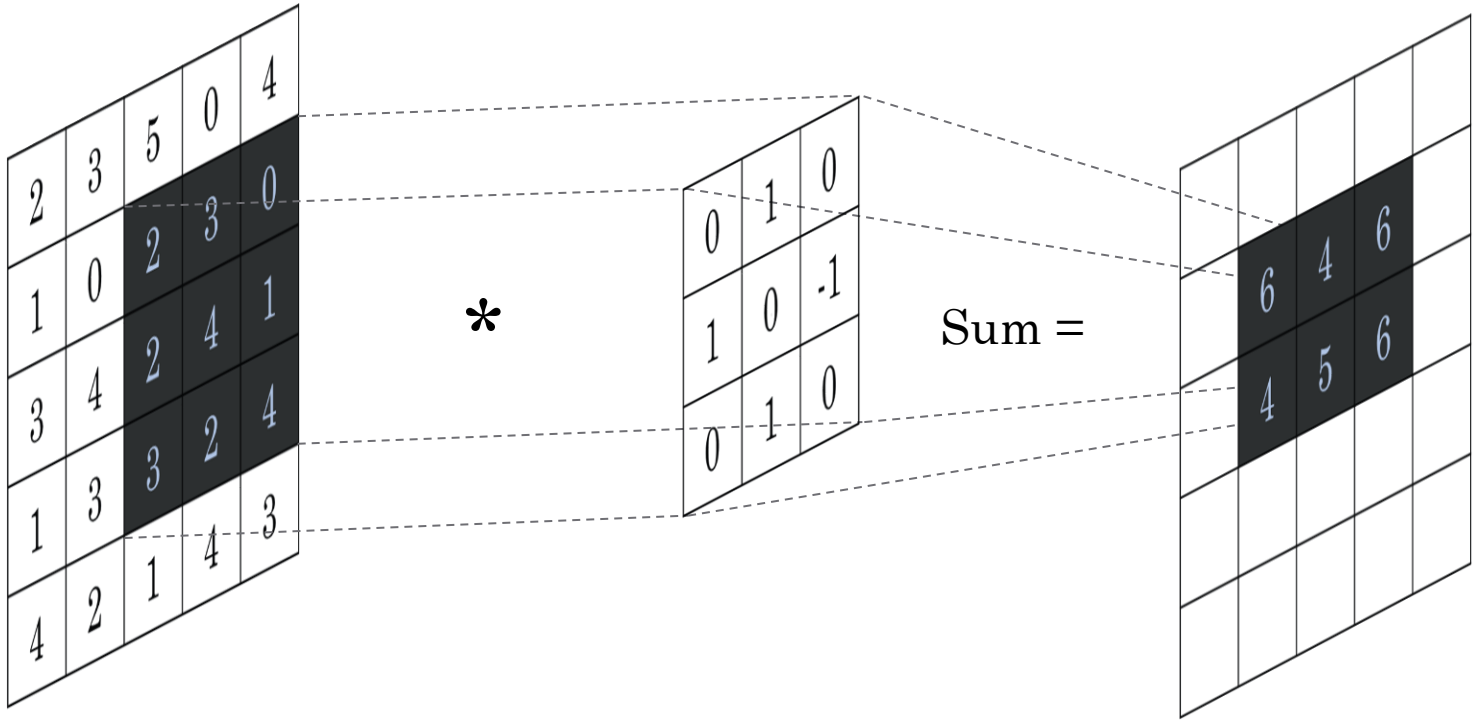
Example of Convolution Operation



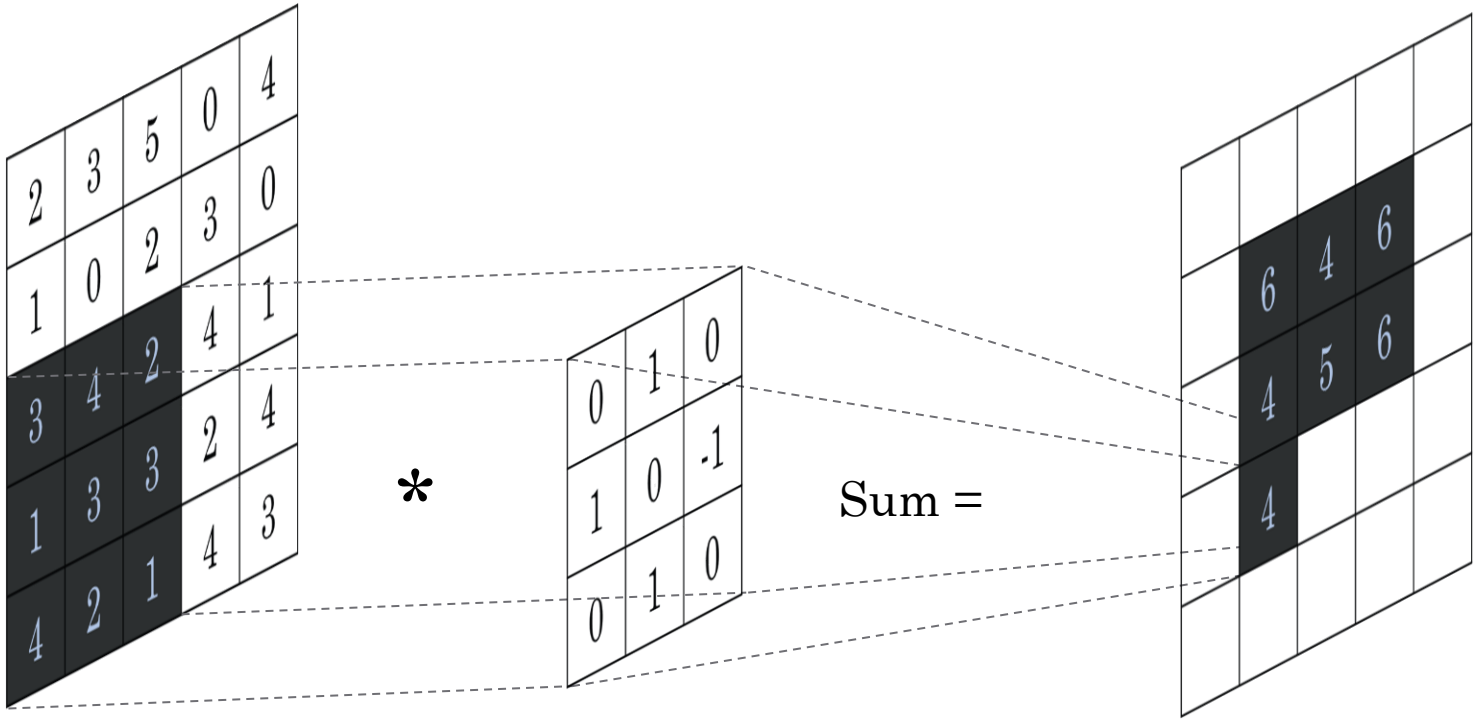
Example of Convolution Operation



Example of Convolution Operation

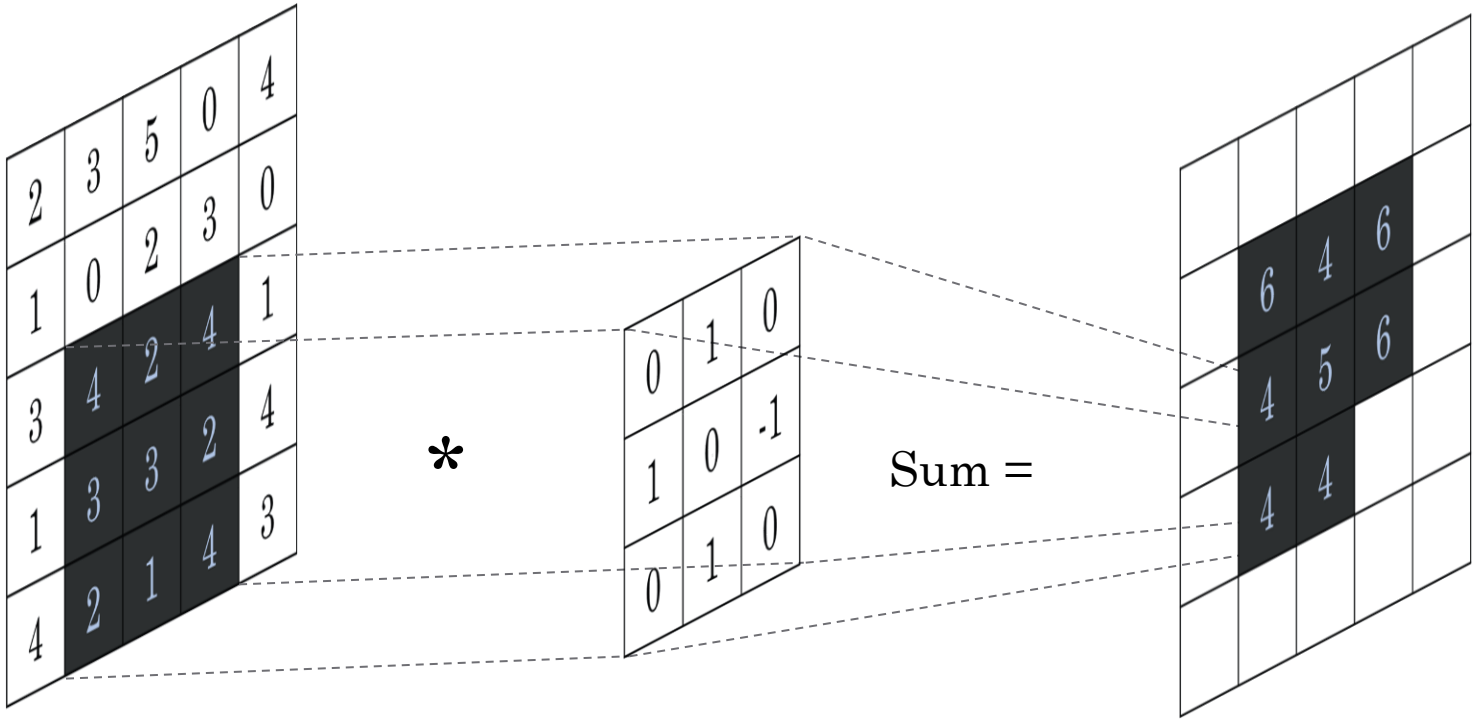


Example of Convolution Operation

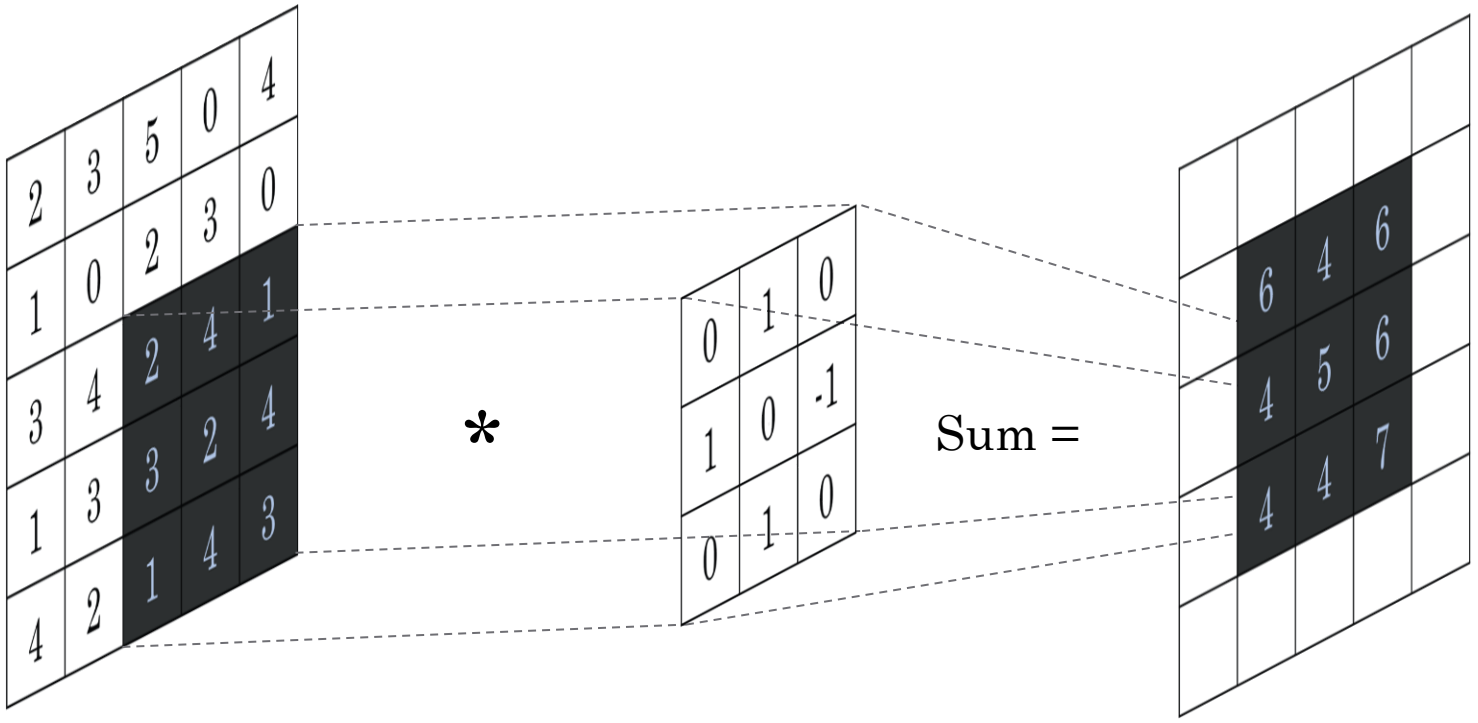


Input Image Filter Output Image

Example of Convolution Operation



Example of Convolution Operation



Input Image Filter Output Image

Convolution as Feature Extractor

We sweep the filter across the entire image to detect these different patterns as

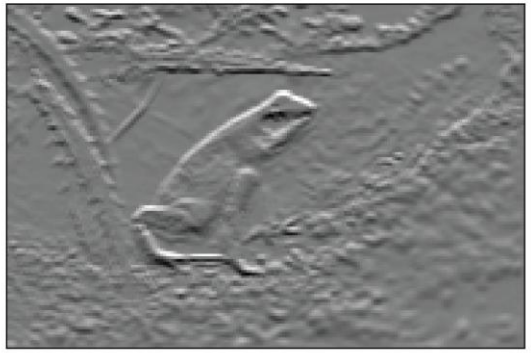
Blur

1	1	1
1	1	1
1	1	1



Horizontal Edges

1	1	1
0	0	0
–1	–1	–1



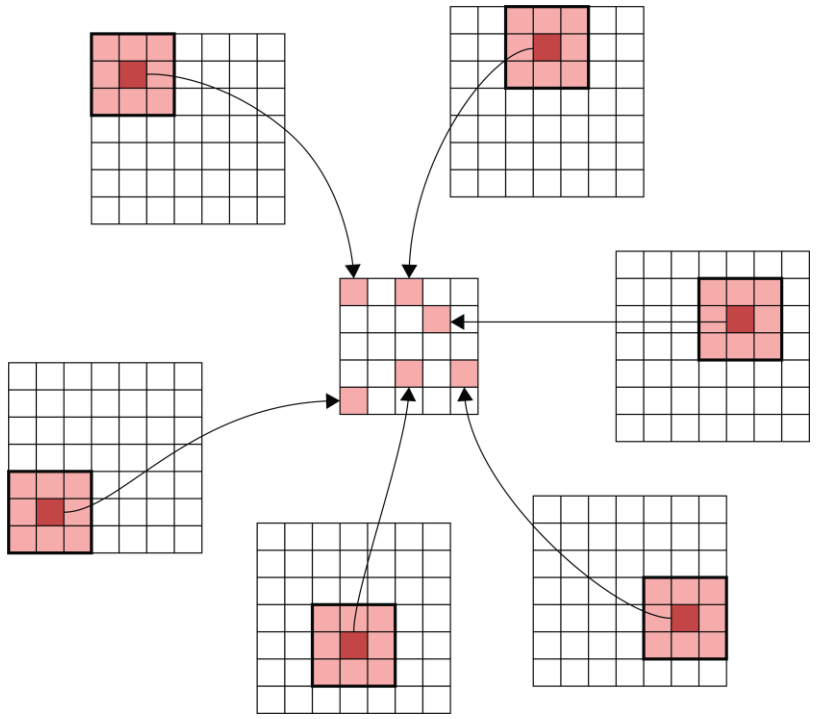
Vertical Edges

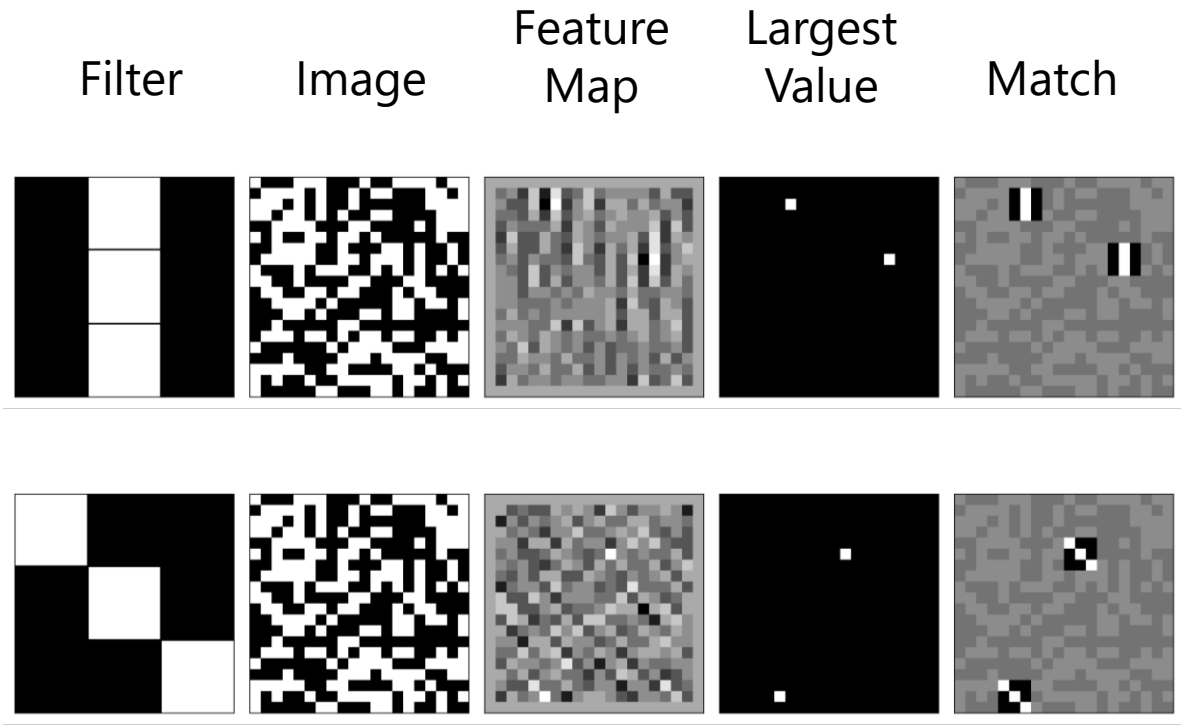
–1	0	1
–1	0	1
–1	0	1



Convolution as Feature Extractor

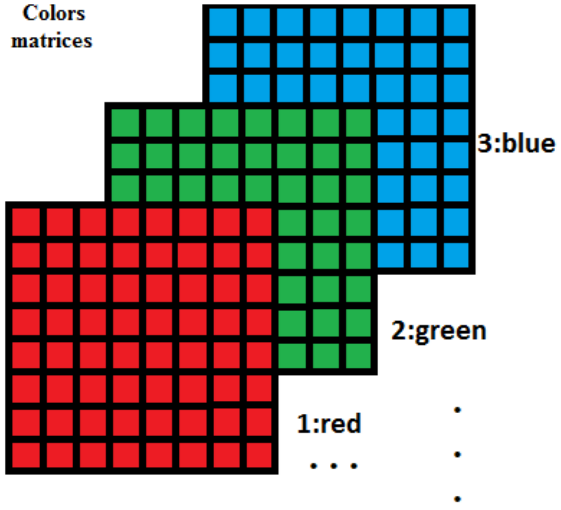
This Process of sweeping the filter across the image and getting a feature map is called convolution



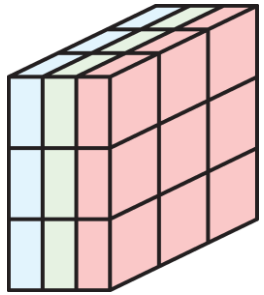


Multichannel Convolution

While performing convolution on input with more than one channel, our filter must also have the same number of channels



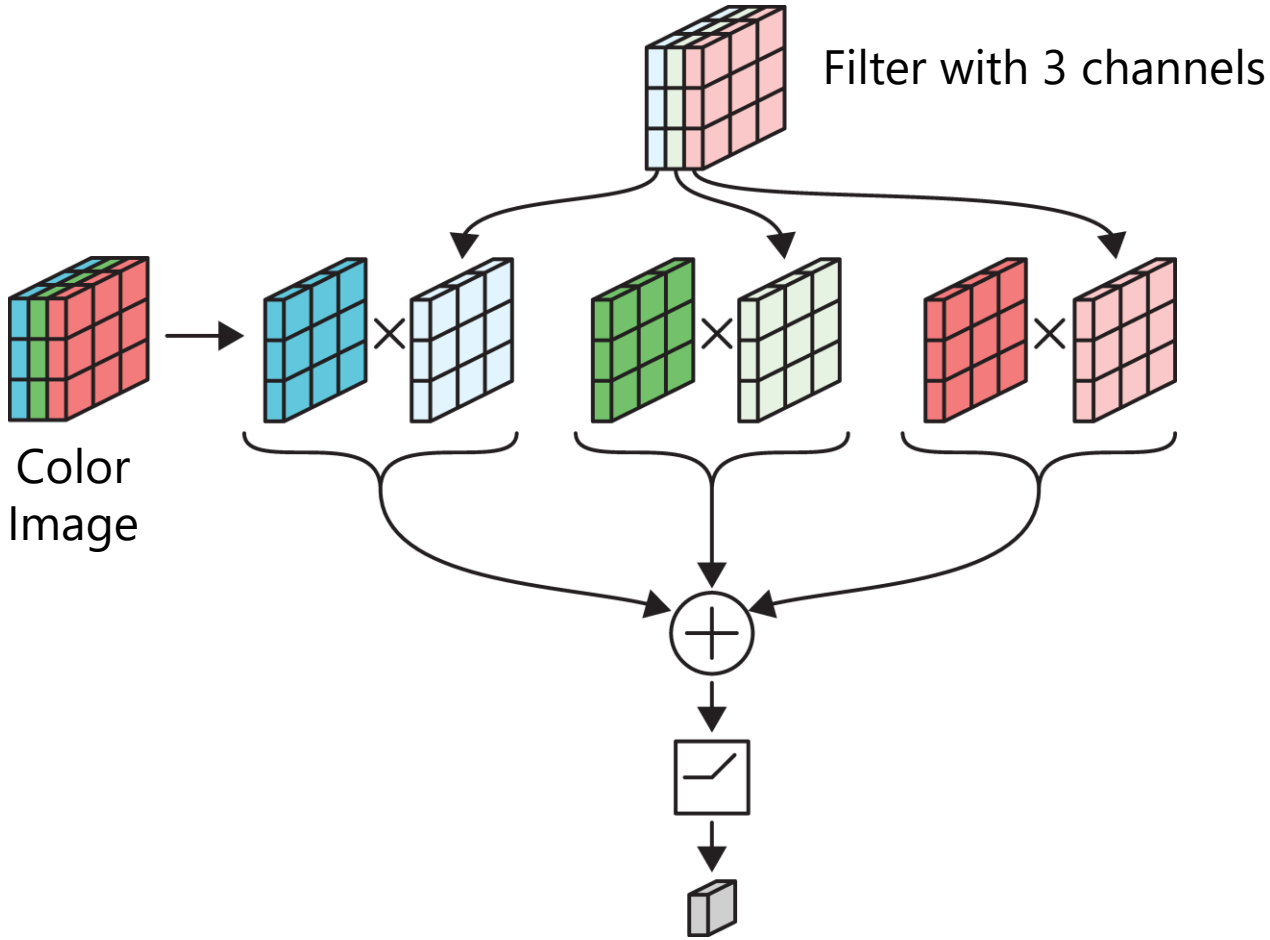
Color Image



Filter with 3 channels

Multichannel Convolution

Each filter will then apply to the corresponding channel and results are added together



Padding in Convolution

- What is padding?
- Why we use padding in **Convolution**?
- In the figure shown, part of the filter 'w' lies outside the image 'f' when processing the edge pixels. As a result, the sum of the products is undefined in that area.
- In the figure shown, the size of the output decreased when we didn't process the edge pixels.
- Padding ensures that the filter can be applied evenly across all pixels.

1	0	-1
1	0	-1
1	0	-1

w

0	4	1	3	0
1	3	1	0	2
4	0	2	4	3
2	0	0	1	4
0	3	1	2	0

f (no padding)

0	4	1	3	0
1	3	1	0	2
4	0	2	4	3
2	0	0	1	4
0	3	1	2	0

⇨

1	0	-1
4	-2	-6
3	-4	-4

Padding in Convolution

Zero Padding

- Zero padding adds zeros around the image border to maintain the same size after filtering.
- **Use cases**: Commonly used in deep learning to ensure output feature maps match input dimensions, especially in multi-layer networks.

Replicate Padding

- Replicate padding copies border pixel values as padding to preserve edge information.
- **Use cases**: Useful in smoothing filters (e.g., Gaussian blur) to avoid artificial edges from zero padding.

1	2	3
4	5	6
7	8	9

f (no padding)

1	0	-1
1	0	-1
1	0	-1

w

0	0	0	0	0
0	1	2	3	0
0	4	5	6	0
0	7	8	9	0
0	0	0	0	0

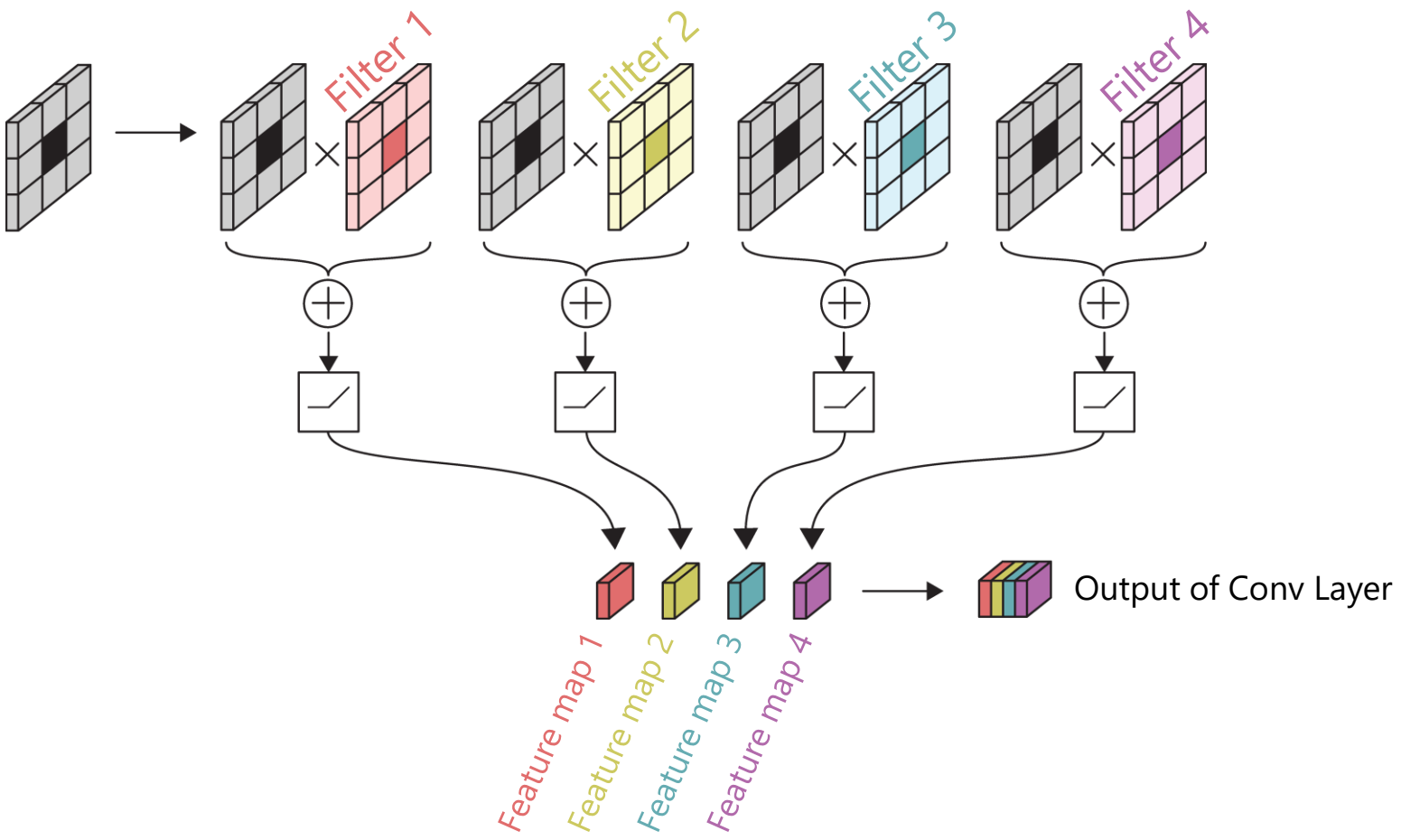
Zero Padding

1	1	2	3	3
1	1	2	3	3
4	4	5	6	6
7	7	8	9	9
7	7	8	9	9

Replicate Padding

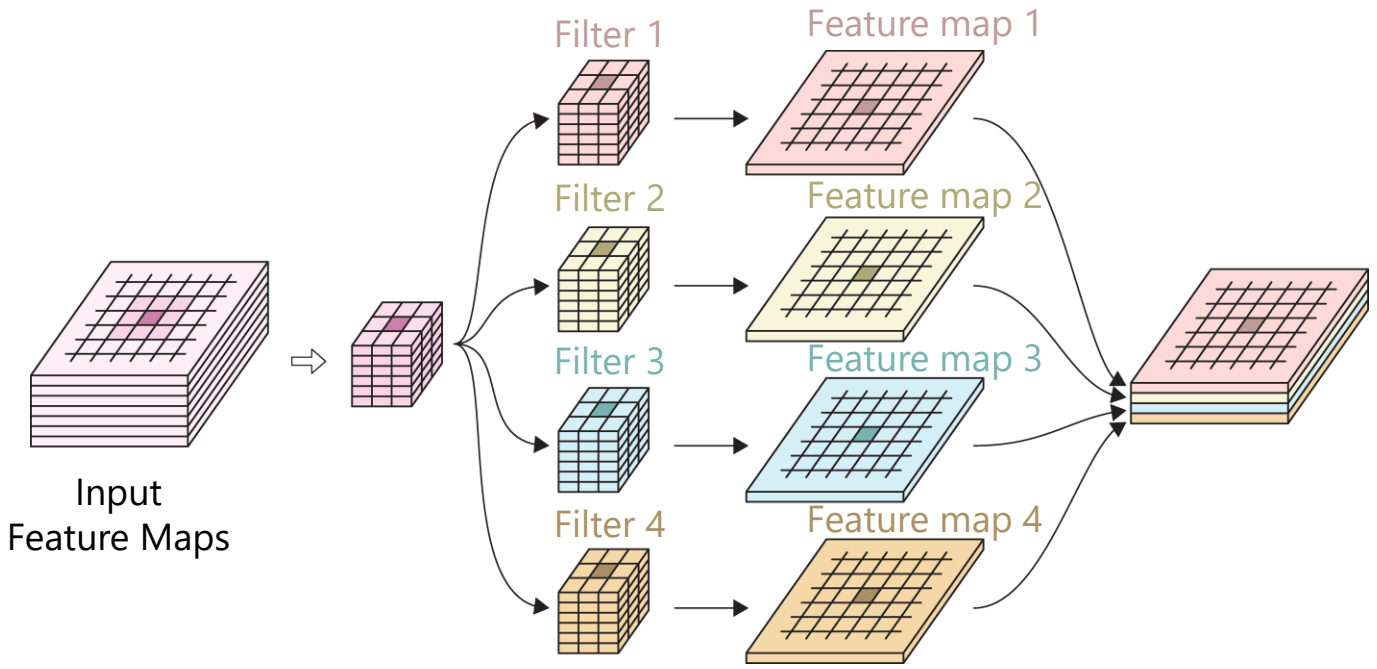
Convolutional Layer

We bundle up multiple filters together, each looking for a different pattern, forming a convolutional layer



Convolutional Layer

A convolutional layer can have many filters, but each one will produce a single feature map



Output of Conv Layer
As many feature maps as there are filters

Pooling Layer

Pooling frees our filters to be precisely in the right place and also reduce the feature map size

3	2	–3	2
1	6	20	5
4	–13	2	6
–2	3	9	3

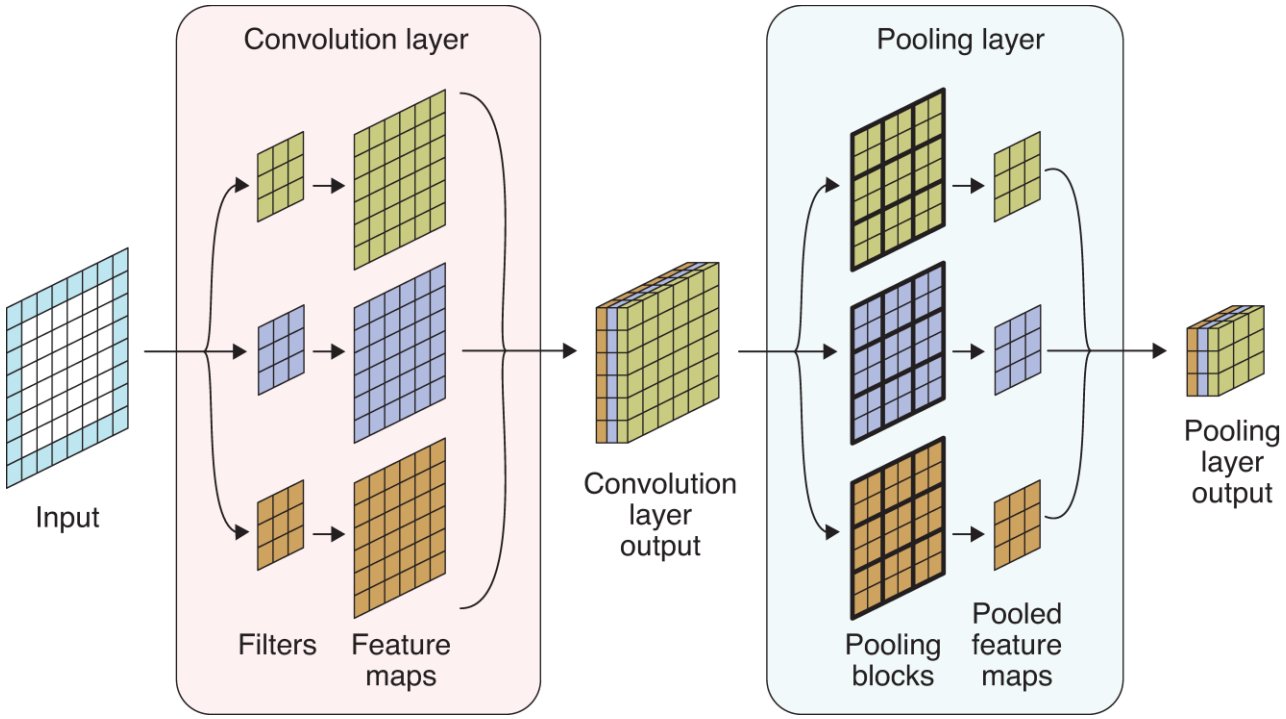
3	2	–3	2
1	6	20	5
4	–13	2	6
–2	3	9	3

3	6
–2	5

Average

6	20
4	9

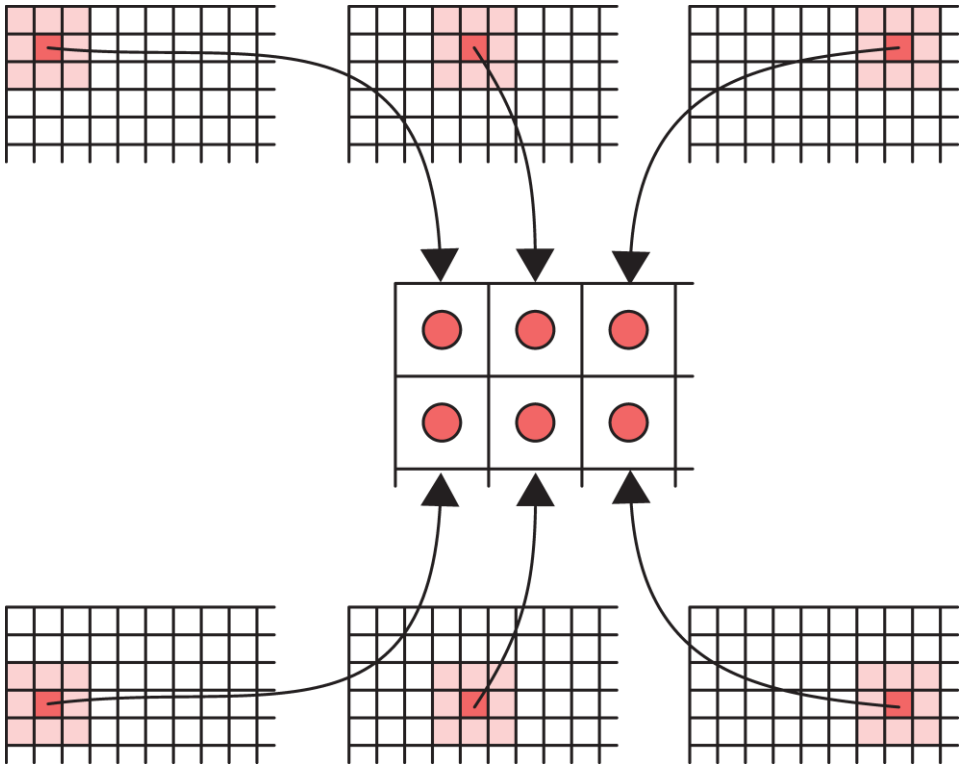
Max



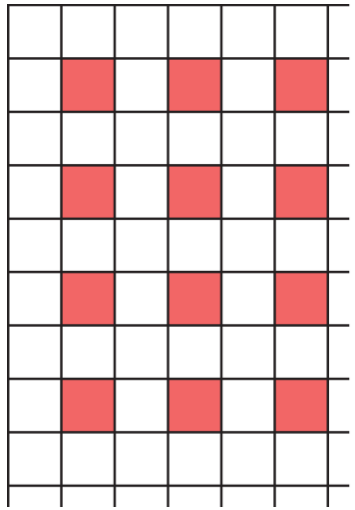
Striding

Instead of sweeping our filters over one pixel at a time, we can move more than one pixel

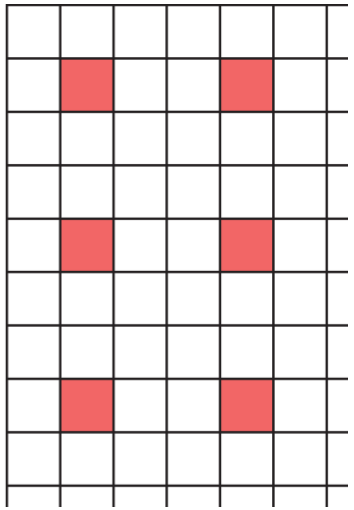
Moving 3 pixels while moving to the right



Moving 2 pixels while moving down



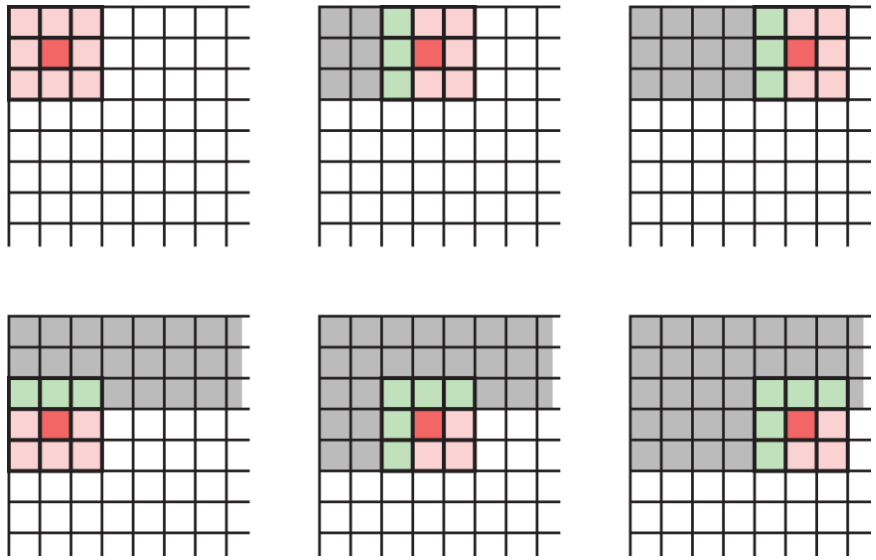
Stride = 2
Center Filter over
every other pixel



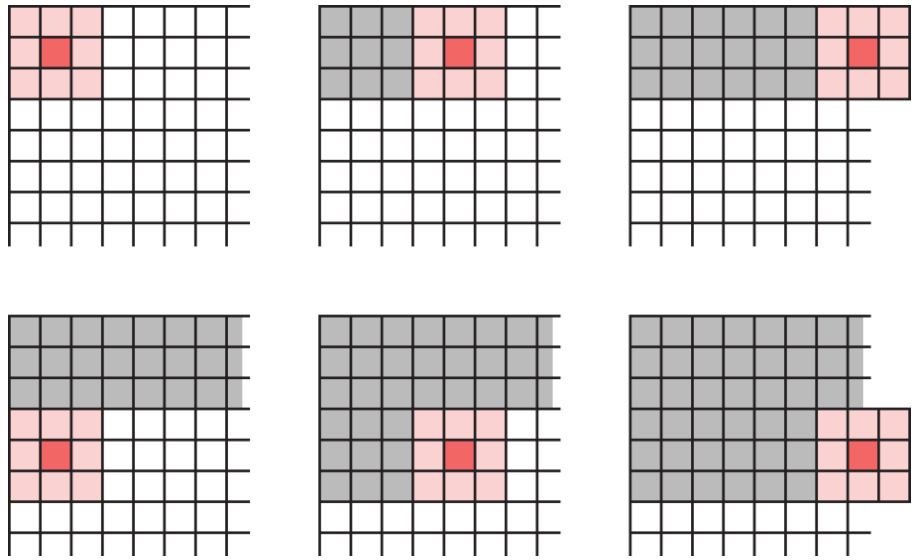
Stride = 3
Center Filter over
every third pixel

Striding

Instead of sweeping our filters over one pixel at a time, we can move more than one pixel



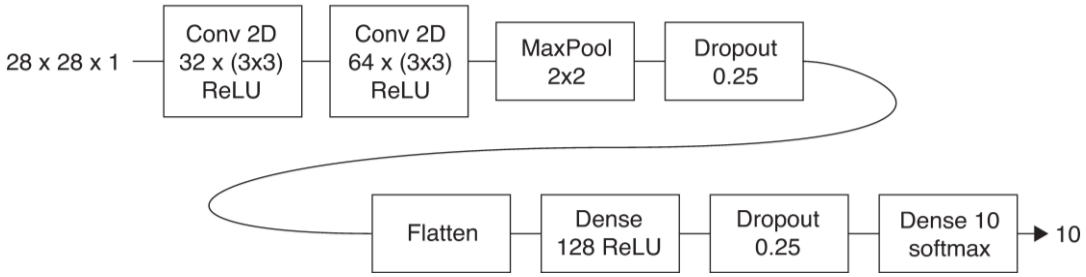
Stride = 2
Center Filter over every other pixel



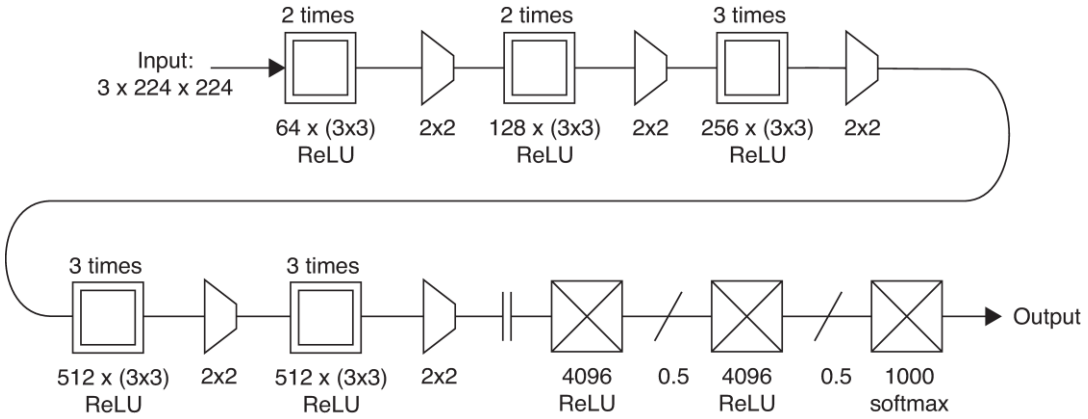
Stride = 3
Center Filter over every third pixel

Convnets in Practice

Convolutional Neural Networks (CNNs) are made of convolutional layers and pooling layers (along with others)



LeNet Architecture



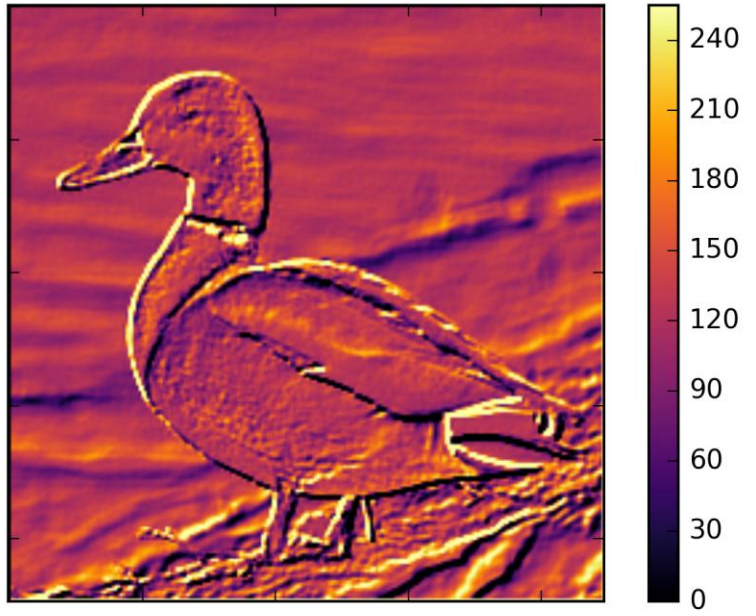
VGG16 Architecture

Visualizing Feature Maps

Let's visualize the feature maps obtained from various CNN blocks of VGG16



Input Image



Feature map of a single filter

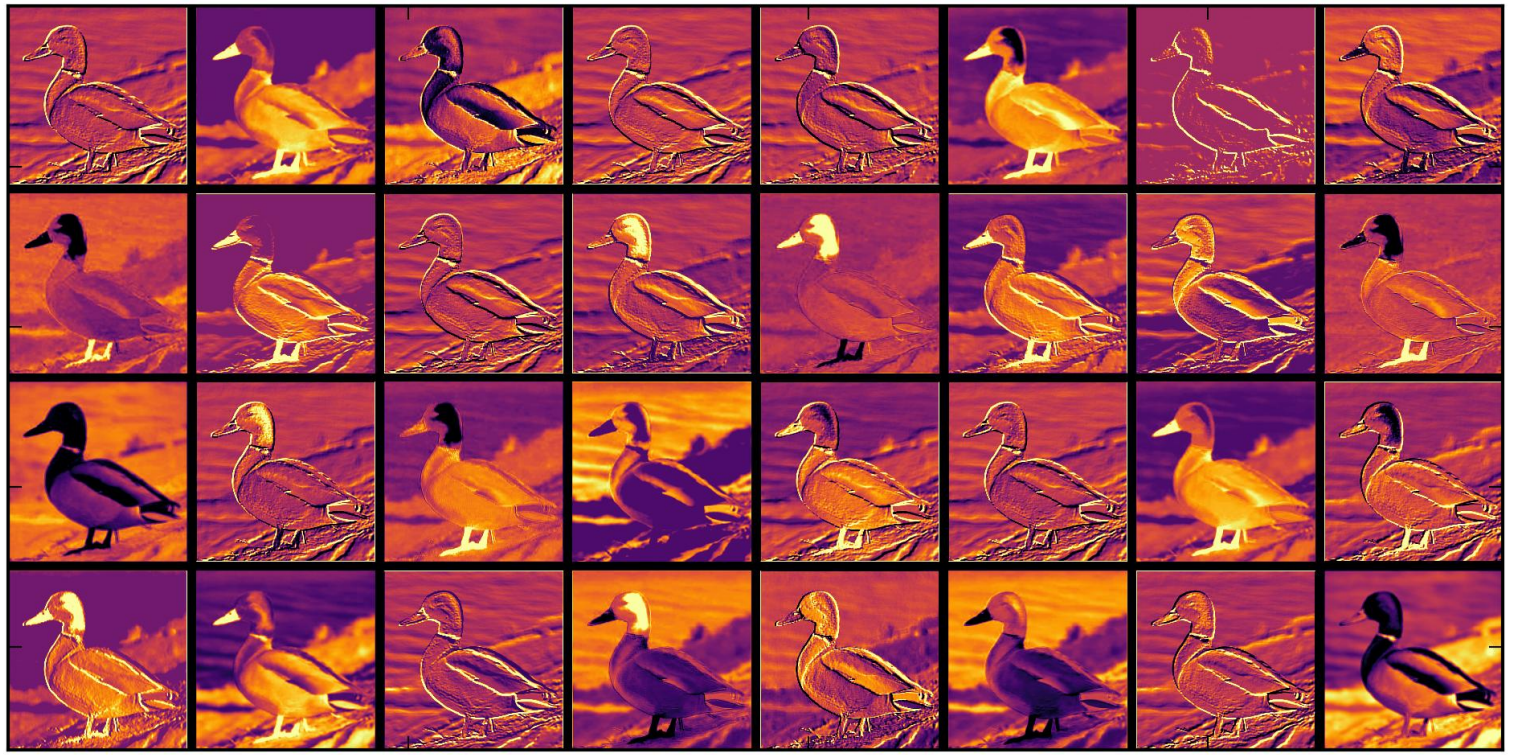
Visualizing Feature Maps

Let's visualize the feature maps obtained from various CNN blocks of VGG16



Input Image

block1_conv1

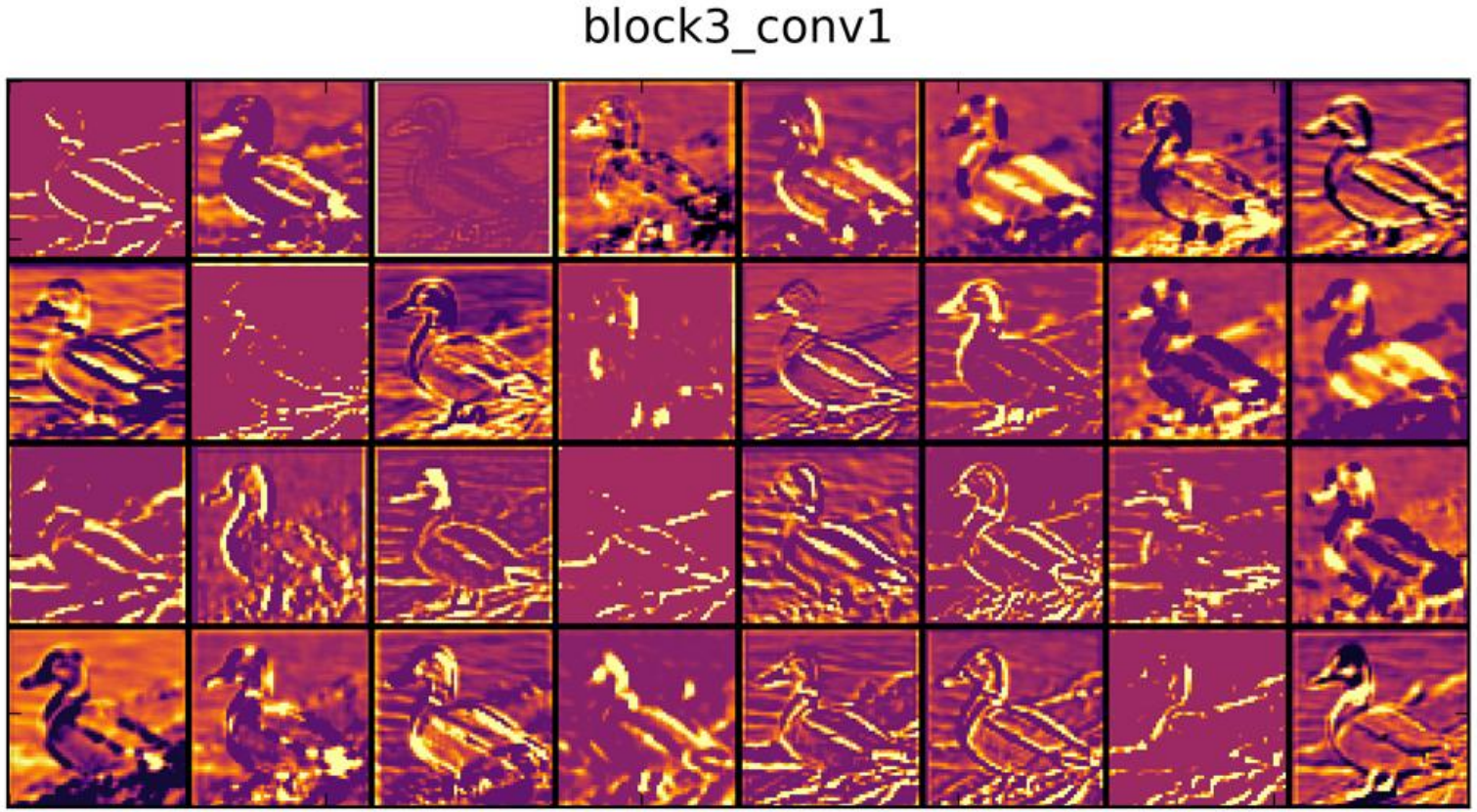


Visualizing Feature Maps

Let's visualize the feature maps obtained from various CNN blocks of VGG16



Input Image

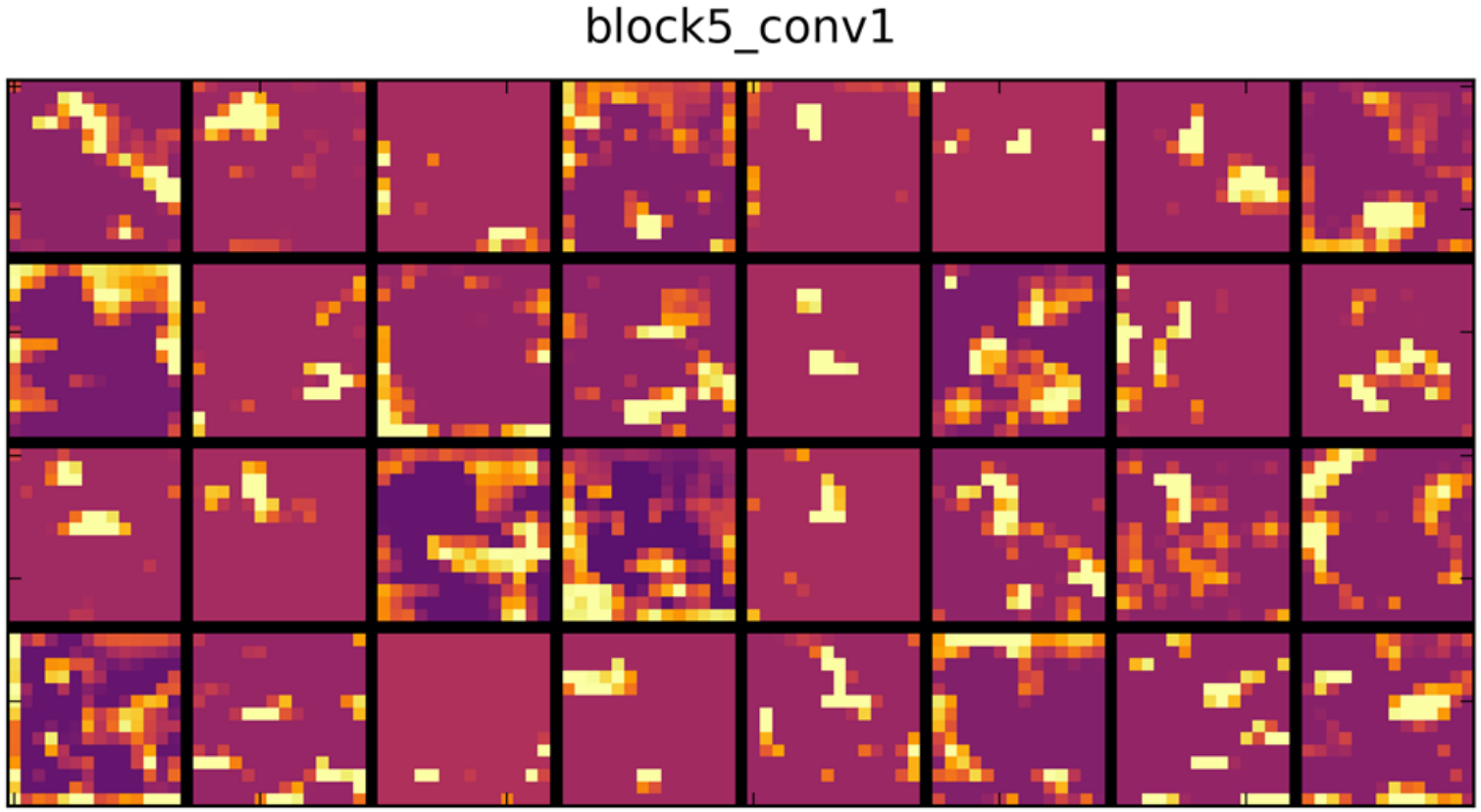


Visualizing Feature Maps

Let's visualize the feature maps obtained from various CNN blocks of VGG16



Input Image



Thank You

